








Modelling Financial Markets

Chapter 4: Portfolio Optimisation



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Why Does Estimating the Optimum Change the Problem?

Easy optimum, hard estimate

- A **portfolio** is a weight vector $\mathrm{w} = (w_1, \ldots, w_N)^\top$ with $\mathrm{w}^\top 1 = 1$
- The optimum is easy to write with known means and covariances, hard to estimate
  - the optimiser loads on the largest estimation errors
  - 16 assets: 16 means + 136 variances and covariances = 152 parameters, estimated from 60 monthly observations
- Can any estimated rule beat equal weights out of sample, after costs?

What the Chapter Contains

Core path

- Mean–variance: global minimum-variance (GMV) and tangency weights, the frontier, constraints
- Estimation risk: why in-sample Sharpe ratios are too optimistic
- Three repairs of the inputs: covariance shrinkage, constraints, Black–Litterman
- Rules that need only Σ: equal risk contribution (ERC) and Hierarchical Risk Parity (HRP)
- Out-of-sample evaluation: turnover and costs, Sharpe-difference tests, Holm and Benjamini–Hochberg
- The application on the Bucharest Stock Exchange (BVB)

Deeper study

- Trading costs of a large investor (case study)
- Research assisted by artificial intelligence (AI) and data snooping
- Appendix: derivations and complete tables
- Textbook: Franke, Härdle & Hafner (2019), Part I

What you should be able to do

- compute minimum-variance, tangency and equal-risk weights
- explain why estimation error favours 1/N
- test a Sharpe difference after costs and multiple testing

Data, Assets and the 1/N Benchmark

Universe	Assets	Returns	Currency
Sectors	9 US sector ETFs	1999/01 – 2026/07	USD
Multi-asset	8 ETFs: equities (3), Treasuries (2), credit (2), gold	2007/05 – 2026/07	USD
Combined	9 sectors + the 7 multi-asset ETFs other than SPY (S&P 500) (16 ETFs)	2007/05 – 2026/07	USD
Romania	8 BVB blue chips; the BET and BET-TR indices	2014/10 – 2026/08	RON

- ETF (exchange-traded fund): a fund traded on an exchange that tracks an index
- Why three universes
  - sectors are similar assets with high correlations
  - multi-asset holds classes with correlations of opposite sign (equities and Treasuries)
  - the combined universe raises N to 16 and amplifies estimation error
- Sample starts: the launch of the sector ETFs (December 1998) and of HYG (April 2007)
- Monthly simple returns from month-end prices, dividends reinvested
  - BET (Bucharest Exchange Trading), the BVB benchmark, is a price index; BET-TR (total return) reinvests dividends
- Excess returns over the one-month US Treasury bill (Kenneth French Data Library)
- **1/N**: equal weights $w_i = 1/N$, rebalanced every month, with no estimation at all
  - the benchmark of every comparison

The Assets in Detail

- **Sectors** (9 US sector ETFs)
  - XLB materials, XLE energy, XLF financials, XLI industrials, XLK technology
  - XLP staples, XLU utilities, XLV health care, XLY discretionary
- **Multi-asset**
  - equities: SPY (S&P 500), EFA (developed markets outside the US and Canada), EEM (emerging markets)
  - bonds: TLT (US Treasuries, 20+ years), IEF (US Treasuries, 7–10 years), LQD (investment grade), HYG (high yield); gold: GLD
- **BVB** (8 blue chips)
  - TLV Banca Transilvania, SNP OMV Petrom, BRD BRD–Groupe Société Générale, TGN Transgaz
  - SNG Romgaz, SNN Nuclearelectrica, EL Electrica, TEL Transelectrica

Notation (1/2)

- **Returns**
  - simple return $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$; $P_{i,t}$: the month-end price of asset i
  - excess return $R^e_{i,t} = R_{i,t} - R_{f,t}$ (in the seminar: r_t and r^e_t); $R_{f,t}$: the risk-free rate
  - log return $\ln(P_{i,t}/P_{i,t-1}) = \ln(1 + R_{i,t})$: adds up over time, not across assets
  - portfolios use simple returns, because $R_{p,t} = \sum_i w_i R_{i,t}$ holds exactly
- **Portfolio** of N assets
  - w: vector of weights; $\boldsymbol{\mu}$: vector of expected (excess) returns; 1: vector of ones
  - $\boldsymbol{\Sigma}$: the $N \times N$ covariance matrix of the returns
  - portfolio mean $\mathrm{w}^\top \boldsymbol{\mu}$, portfolio variance $\mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}$

Notation (2/2)

- **Sharpe ratio** (Sharpe (1966)): $SR_p = \mathbb{E}[R_p^e]/\sigma(R_p^e)$
  - the mean excess return per unit of volatility σ
  - monthly figures annualised with $\times 12$ (mean) and $\times\sqrt{12}$ (volatility, Sharpe)
- **Maximum drawdown**: $\min_t(W_t/\max_{s\le t} W_s - 1)$
  - the largest fall of wealth W_t from a previous peak
- **Out of sample** (OOS): the weights held in month t use only data up to month $t-1$
- **In sample**: the same data serve for estimation and for evaluation
- **Rules of this chapter**
  - MV (mean–variance): tangency portfolio from sample moments; GMV: global minimum variance
  - MV-LO and GMV-LO: their long-only (LO) versions; GMV-LW: GMV with Ledoit–Wolf (LW) covariance shrinkage
  - ERC (equal risk contribution) and HRP: risk-based rules; 1/N: equal weights

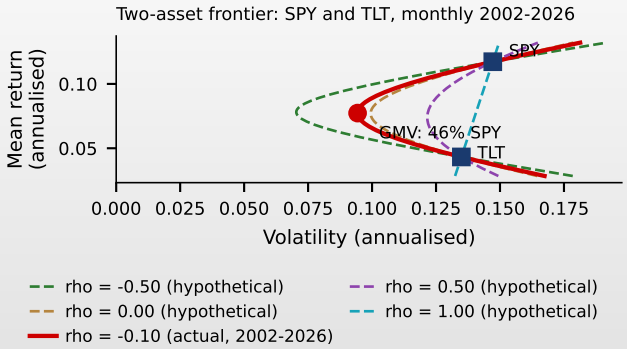
The Markowitz Problem

- The question of this section: which weights do we choose if $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are known?
- Markowitz (1952): investors care about the mean and the variance of portfolio returns
- **Minimum-variance frontier**: minimum variance for every target mean m
  - its upper branch, $m \geq \mu_{\text{GMV}}$ (the mean of the GMV portfolio), is the **efficient frontier**

$$\min_{\mathrm{w}} \ \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w} \quad \text{s.t.} \quad \mathrm{w}^\top \boldsymbol{\mu} = m, \quad \mathrm{w}^\top 1 = 1$$

  - constraints: the target mean m and full investment (weights sum to 1)
- Utility form, risk aversion $\gamma > 0$: $\max_{\mathrm{w}} \ \mathrm{w}^\top \boldsymbol{\mu} - \frac{\gamma}{2} \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}$ s.t. $\mathrm{w}^\top 1 = 1$
  - as γ varies, the solution traces exactly the efficient branch
- Weights may be negative: a **short sale** (selling a borrowed asset)

Two Assets: How Correlation Diversifies



- Two assets, correlation ρ: $\sigma_p^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$
  - w: weight of asset 1, σ_1, σ_2: volatilities, σ_p: portfolio volatility
- Minimised at $w^* = \dfrac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$
- SPY and TLT, equities and long Treasuries, monthly 2002–2026: $\rho = -0.10$
  - $w^* = 46\%$ SPY, volatility 9.4% (SPY 14.7%, TLT 13.5%)
  - time-varying correlation: Chapter 6
- Chart: means of total returns and annualised volatilities, in decimals ($0.10 = 10\%$)
- The lower ρ, the more the curve bends to the left

MFM_ch4_two_asset

The Global Minimum-Variance (GMV) Portfolio

- The problem: $\min_{\mathrm{w}} \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}$ s.t. $\mathrm{w}^\top 1 = 1$, $\boldsymbol{\Sigma}$ positive definite
- Lagrangian $\mathcal{L} = \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w} - \lambda(\mathrm{w}^\top 1 - 1)$
  - λ: the Lagrange multiplier of the budget constraint
- First-order condition $2\boldsymbol{\Sigma}\mathrm{w} = \lambda 1$, so $\mathrm{w} = \frac{\lambda}{2}\boldsymbol{\Sigma}^{-1}1$
- The budget gives $\frac{\lambda}{2} = 1/A$, with $A = 1^\top \boldsymbol{\Sigma}^{-1} 1$
- Result: $\mathrm{w}_{\mathrm{GMV}} = \boldsymbol{\Sigma}^{-1}1/A$, variance $\sigma^2_{\mathrm{GMV}} = 1/A$
- Needs only $\boldsymbol{\Sigma}$, not the means

The Tangency Portfolio

- The problem: $\max_{\mathrm{w}} \mathrm{w}^\top \boldsymbol{\mu} / \sqrt{\mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}}$ s.t. $\mathrm{w}^\top 1 = 1$
  - $\boldsymbol{\mu}$: the expected **excess** returns
- First-order condition: $\boldsymbol{\mu} = \dfrac{\mathrm{w}^\top \boldsymbol{\mu}}{\mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}} \boldsymbol{\Sigma} \mathrm{w}$, so $\mathrm{w} \propto \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$
- The budget gives $\mathrm{w}_{\mathrm{MV}} = \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} / B$, with $B = 1^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$
- Maximum Sharpe ratio θ: $\theta^2 = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$
  - needs $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$
- The Sharpe ratio of w_{MV} is $\mathrm{sign}(B)\,\theta$: the maximum only if $B > 0$
- $B < 0$ gives the minimum-Sharpe portfolio
  - for $B = 0$ there is no fully invested solution

Separation and the Frontier in Closed Form



James Tobin, Nobel Prize in Economics 1981

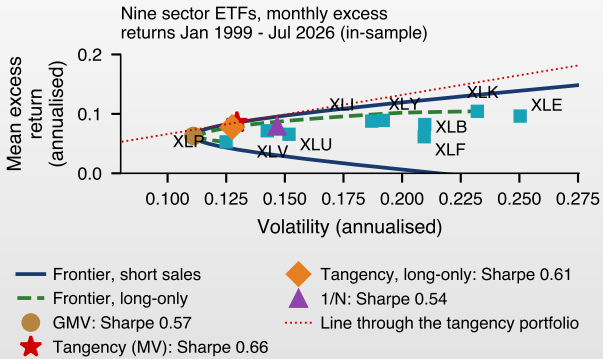
Photo: unknown author (1962); public domain; Wikimedia Commons

- Tobin (1958): add a riskless asset paying R_f
  - every investor holds the **same** risky portfolio, the tangency portfolio, mixed with cash or borrowing
  - risk aversion changes only the mix: **two-fund separation**, the bridge to the Capital Asset Pricing Model (CAPM; Chapter 3)
- Merton (1972): with short sales, the frontier is a hyperbola
  - Σ positive definite, μ not proportional to 1
  - A and B as on the two previous slides; $C = \mu^\top \Sigma^{-1} \mu = \theta^2$, $D = AC - B^2 > 0$

$$\sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}$$

- $\sigma^2(m)$: the smallest variance with mean m
  - a parabola in the (σ^2, m) plane (derivation: ▸ Appendix)
  - the GMV sits at $m = B/A$, $\sigma^2 = 1/A$
  - every frontier portfolio mixes any two distinct frontier portfolios, e.g. GMV and tangency (if $B \neq 0$)

The Frontier of US Sectors



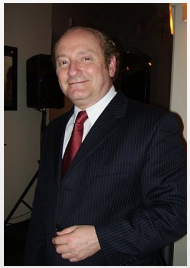
- Monthly excess returns, January 1999 – July 2026 ($T = 331$)
  - axes annualised, in decimals
  - all parameters estimated on the full sample: an **in-sample** picture
- Tangency: Sharpe 0.66; GMV: 0.57 (volatility 11.1%); 1/N: 0.54
- Tangency: short positions (-0.37 financials, -0.21 materials)
  - the largest long position in health care ($+0.37$)
- Long-only tangency: Sharpe 0.61
  - materials, financials and industrials get exactly zero, a **corner solution**
- Dotted line: mixes of the tangency portfolio with cash, from the origin, slope 0.66
- The gain over 1/N is small even with perfect hindsight

MFM_ch4_frontier

Constraints

- **Long-only** (LO): $w_i \geq 0$
  - no closed form, solved by quadratic programming
- Position limits: $w_i \leq 0.2$
  - sector or asset-class limits: $\sum_{i \in G} w_i \leq c_G$
  - G a group of assets, c_G its cap
- Leverage limit: $\sum_i |w_i| \leq L$
  - $\sum_i |w_i|$: gross exposure
  - $L = 1$ means no short sales
- Frost & Savarino (1988): constraining weights improved out-of-sample performance
  - the reason: constraints act as shrinkage of the covariance matrix (Jagannathan & Ma (2003); slide Constraints Are Shrinkage Too)

Means Are Hard to Estimate: Merton's Argument

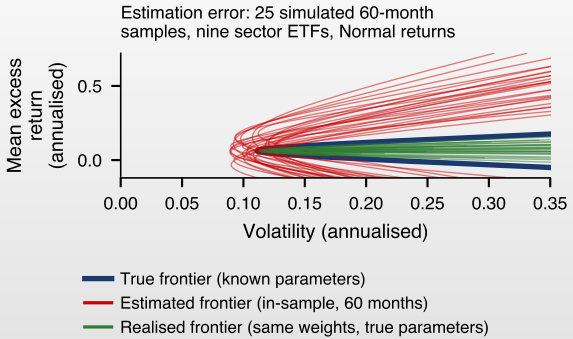


Robert C. Merton, Nobel Prize in Economics 1997

Photo: Digarnick (2006); public domain; Wikimedia Commons

- The question of this section: how much does the optimal portfolio lose when $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are replaced by estimates?
- Merton (1980): Y years sampled h times a year
  - i.i.d. Normal returns (a diffusion) with annual mean μ and variance σ^2
  - $\mathrm{Var}(\hat{\mu}) = \sigma^2/Y$: independent of h; only the **span** of the sample helps
  - $\mathrm{Var}(\hat{\sigma}^2) \approx 2\sigma^4/(hY) \to 0$ as $h \to \infty$ (Normal; in general $(\kappa - 1)\sigma^4/(hY)$, κ the kurtosis)
- SPY, $Y = 24$, $\sigma = 14.7\%$: standard error 3.0%
  - 95% interval $[5.8\%, 17.6\%]$ for a mean of 11.7%
- A standard error of 1% needs $(0.147/0.01)^2 \approx 217$ years
- Chopra & Ziemba (1993): errors in means cost about ten times more than errors in variances
  - rules without means (GMV, risk parity) are more robust

Error Maximisation: Estimated versus Realised Frontiers (1/2)



- Michaud (1989): optimisers are **estimation-error maximisers**
  - Best & Grauer (1991): a tiny change in one mean moves many weights
- True parameters = sector moments 1999–2026
  - draw 60 Normal months, optimise, evaluate with the true parameters
- Chart: 25 simulated 60-month samples
  - red: estimated frontiers, too optimistic
  - green: the same portfolios under the true parameters, all below the true frontier

MFM_ch4_estimation_error

Error Maximisation: Estimated versus Realised Frontiers (2/2)

- 2000 simulated tangency portfolios: median estimated maximum Sharpe 1.56
  - median true Sharpe 0.30, true optimum 0.66
- 96% have a true Sharpe below 1/N (0.54), 7.5% below zero
- In 11% of samples $\hat{B} = 1^\top \hat{\Sigma}^{-1} \hat{\mu} < 0$
  - so the tangency formula returns the minimum-Sharpe portfolio

MFM_ch4_estimation_error

The Plug-in Sharpe Ratio Is Biased (1/2)

- i.i.d. $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ returns, T months, N assets (Kan & Zhou (2007))
  - $T\hat{\boldsymbol{\Sigma}}$: Wishart distribution, Appendix
  - $\hat{\theta}^2 = \hat{\boldsymbol{\mu}}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}$: the in-sample squared maximum Sharpe ratio, from the maximum-likelihood $\hat{\boldsymbol{\mu}}$, $\hat{\boldsymbol{\Sigma}}$; θ^2: its true value

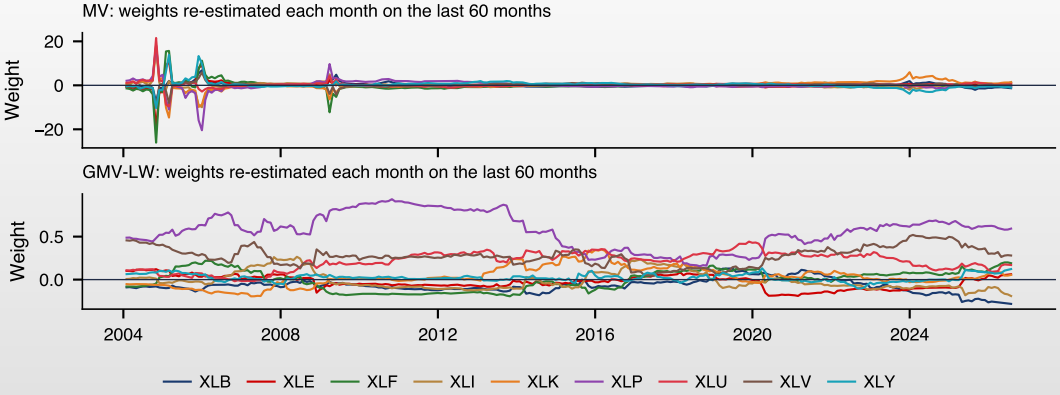
$$\mathbb{E}[\hat{\theta}^2] = \frac{T}{T-N-2}\left(\theta^2 + \frac{N}{T}\right) = \frac{T\theta^2 + N}{T-N-2}, \qquad T > N+2$$

- Sectors: $N = 9$, $T = 60$, true $\theta = 0.66$ a year: $\sqrt{12\,\mathbb{E}[\hat{\theta}^2]} = 1.65$ a year
  - a root mean square, not a median
  - the simulated median 1.56 is lower: $\hat{\theta}$ is right-skewed; derivation: Seminar 4, A8

The Plug-in Sharpe Ratio Is Biased (2/2)

- Unbiased estimator of θ^2: $\hat{\theta}_u^2 = \left[(T - N - 2)\,\hat{\theta}^2 - N\right]/T$
  - It can be negative; truncating at zero gives a nonnegative but upward-biased estimator, and $\sqrt{\hat{\theta}_u^2}$ is not unbiased for θ
- Sectors, full sample ($T = 331$): plug-in $\hat{\theta} = 0.66$, bias-corrected $\sqrt{\hat{\theta}_u^2} = 0.31$ a year
  - more than half of the in-sample Sharpe is noise, even with 27 years
- Tangency weights: $\mathrm{w} = \mathrm{b}/1^\top \mathrm{b}$ (Britten-Jones (1999))
  - b: the ordinary least squares (OLS) coefficients of a regression of 1 on the excess returns, without a constant
  - derivation: Appendix; Seminar 4, A9 and B6

Unstable Weights (1/2)



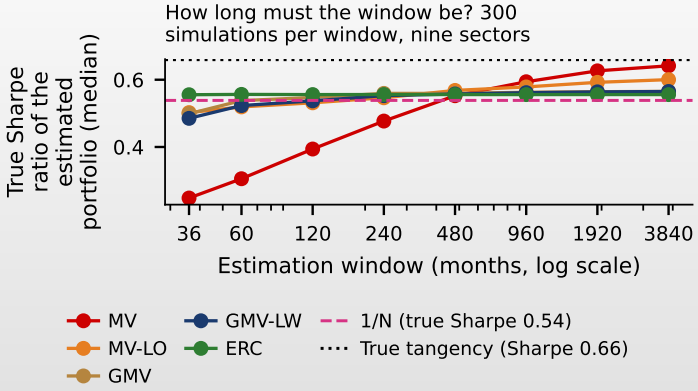
MFM_ch4_oos_backtest

Unstable Weights (2/2)

- Nine sector ETFs; a 60-month rolling window
  - one of the two windows of DeMiguel, Garlappi & Uppal (2009) (60 and 120 months)
  - a longer window cuts estimation error, a shorter one tracks regime changes faster
- MV: median gross exposure $\sum_i |w_i| = 4.9$
  - above 3 in 90% of the months
  - largest single weight 26 in absolute value
- Explosions in 2004–2006 and 2009
  - the denominator $1^\top \hat{\Sigma}^{-1} \hat{\mu}$ is close to zero, so the tangency weights have very heavy tails
  - Negative in 12 of 271 windows: MV then holds the minimum-Sharpe portfolio
- GMV-LW (bottom): gross exposure 1.5, largest weight 0.94
- Unstable weights mean high turnover, hence high trading costs

MFM_ch4_oos_backtest

How Long Must the Window Be to Beat 1/N? (1/2)



MFM_ch4_window_length

How Long Must the Window Be to Beat 1/N? (2/2)

- DeMiguel, Garlappi & Uppal (2009): 14 models on 7 data sets
  - none beats 1/N consistently in Sharpe ratio, certainty-equivalent return or turnover
  - certainty-equivalent return: mean $-\frac{\gamma}{2}\times$ variance, the riskless return worth the same utility
- In their calibration MV needs a window of about 3000 months for 25 assets
- Chart: 300 simulated samples per window
  - horizontal axis: window, months, log scale
  - population parameters = sector moments
  - vertical axis: the median true Sharpe ratio
- MV beats 1/N only with 480 months (40 years)
  - median true Sharpe 0.55 vs 0.54
  - with 60 months 0.31
- MV-LO, GMV, GMV-LW are close to 1/N from 60 months on
  - ERC is almost insensitive, 0.56
- In this calibration 1/N is nearly efficient (true tangency 0.66)
  - simulation evidence, not an empirical test

MFM_ch4_window_length

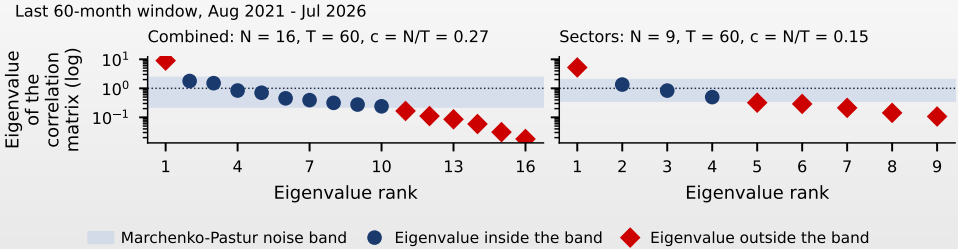
Why the Sample Covariance Matrix Fails

- The question of this section: how do we estimate $\boldsymbol{\Sigma}$ so that $\hat{\boldsymbol{\Sigma}}^{-1}$ does not amplify noise?
- Sample eigenvalues are too dispersed: the largest too large, the smallest too small
- **Condition number**: largest over smallest eigenvalue
  - a large value means $\boldsymbol{\Sigma}^{-1}$ amplifies noise
- Optimisers invert $\boldsymbol{\Sigma}$, so they load on the directions with the smallest estimated variance
- The demeaned sample matrix has rank at most $T - 1$: singular, not invertible, whenever $T \leq N$
- **Marchenko–Pastur** benchmark (Marčenko & Pastur (1967))
  - i.i.d. returns with $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ ($\mathbf{I}$: identity matrix)
  - as $N, T \to \infty$ with $N/T \to c \in (0, 1)$, the sample eigenvalues fill

$$[\lambda_-, \lambda_+] = \left[\sigma^2(1 - \sqrt{c})^2,\ \sigma^2(1 + \sqrt{c})^2\right]$$

  - λ_-, λ_+: the smallest and largest eigenvalue that pure noise produces
  - $c = 16/60 = 0.27$: band $[0.23, 2.30]$ for a correlation matrix ($\sigma^2 = 1$), a ten-fold spread from noise alone
  - Laloux et al. (1999): eigenvalues outside the band are candidates for structure; those inside cannot be told from noise by this comparison alone

Sample Eigenvalues versus the Noise Band (1/2)



- Correlation matrices of monthly excess returns, last 60-month window (August 2021 – July 2026)
  - shaded band: $[\lambda_-, \lambda_+]$ for $\sigma^2 = 1$
- Only the market mode lies above λ_+
  - 9.0 (56% of the trace) for 16 ETFs, 5.2 for nine sectors ($\lambda_+ = 1.92$)
  - with the noise variance rescaled to $\sigma^2 = 1 - \lambda_1/N$ (Laloux et al. (1999)), three eigenvalues exceed the band in each universe

MFM_ch4_inference

Sample Eigenvalues versus the Noise Band (2/2)

- Six of 16 eigenvalues lie **below** the band: near-riskless combinations
  - such as long TLT and short IEF, scaled by their volatilities
- GMV puts large weights on these combinations
  - if their variance is underestimated, so is the GMV risk

MFM_ch4_inference

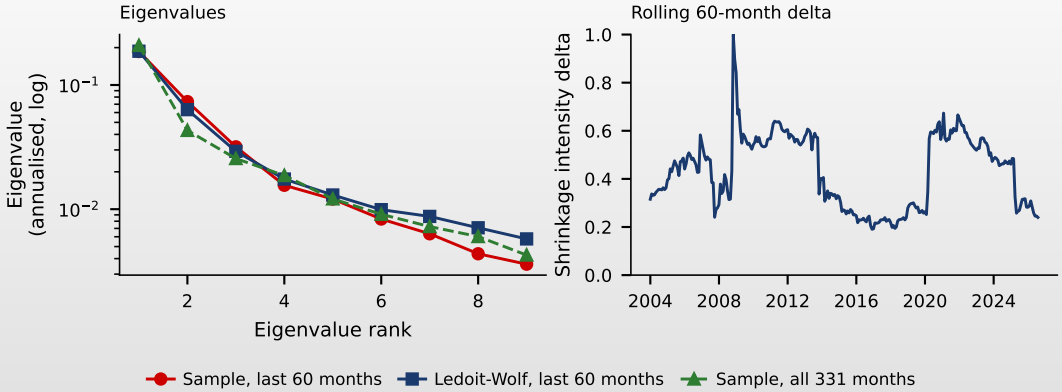
The Shrinkage Estimator

- Shrink the sample matrix S towards a structured target F

$$\hat{\boldsymbol{\Sigma}}_{\mathsf{LW}} = \delta\,\mathsf{F} + (1-\delta)\,\mathsf{S}, \qquad 0 \le \delta \le 1$$

- Possible targets
  - the single-index model (Ledoit & Wolf (2003))
  - the scaled identity (Ledoit & Wolf (2004a))
  - constant correlation (Ledoit & Wolf (2004b))
- The target used here: $f_{ii} = s_{ii}$, $f_{ij} = \bar{r}\sqrt{s_{ii}s_{jj}}$ for $i \neq j$
  - $\bar{r}$: the average sample correlation
- Why this target: the sectors are all US equities with similar correlations
  - the target needs no choice of a market index
- Optimal intensity: $\delta^* = \max\{0, \min\{1, \hat{\kappa}/T\}\}$, $\hat{\kappa} = (\hat{\pi} - \hat{\rho})/\hat{\gamma}$
  - $\hat{\pi} - \hat{\rho}$: the estimated noise of S
  - $\hat{\gamma} = \|\mathsf{F} - \mathsf{S}\|^2$: the distance to the target (not the risk aversion γ)
- δ^* falls like $1/T$: in long samples $\delta^* \to 0$ (Appendix)

Shrinkage on US Sectors (1/2)



MFM_ch4_shrinkage

Shrinkage on US Sectors (2/2)

- Nine sector ETFs, monthly excess returns
  - left: annualised eigenvalues (log scale)
  - right: δ^* on rolling 60-month windows
- Last window, August 2021 – July 2026: $\delta^* = 0.24$
- The smallest eigenvalues rise
  - the largest is almost unchanged, the second falls from 0.073 to 0.063
- Condition number $51.6 \to 32.5$
  - the matrix on all 331 months (green) has 48.7
- Rolling δ^*: mean 0.43, range 0.19–1.00
- $\delta^* = 1$ only in the window ending in October 2008
  - the crisis month triples the estimated noise $\hat{\pi}$ of S, while the distance $\hat{\gamma}$ to the target barely moves

MFM_ch4_shrinkage

Constraints Are Shrinkage Too

- Jagannathan & Ma (2003): the long-only GMV equals the unconstrained GMV of a modified covariance matrix

$$\tilde{S} = S - (\boldsymbol{\lambda} 1^\top + 1\boldsymbol{\lambda}^\top), \qquad \mathcal{L} = \tfrac{1}{2} w^\top S w - \lambda_0 (w^\top 1 - 1) - \boldsymbol{\lambda}^\top w$$

  - $\boldsymbol{\lambda} \geq 0$: the Lagrange multipliers of the constraints $w_i \geq 0$ in $\mathcal{L}$; $\lambda_i > 0$ only where the constraint binds
  - The first-order condition $Sw = \lambda_0 1 + \boldsymbol{\lambda}$ and $\boldsymbol{\lambda}^\top w = 0$ give $\tilde{S}w = \lambda_0 1$: the GMV condition for $\tilde{S}$
  - step by step, with the budget $1^\top w = 1$: $\tilde{S}w = Sw - \boldsymbol{\lambda}(1^\top w) - 1(\boldsymbol{\lambda}^\top w) = \lambda_0 1 + \boldsymbol{\lambda} - \boldsymbol{\lambda} - 0$
- $\tilde{s}_{ij} = s_{ij} - \lambda_i - \lambda_j$: a binding constraint on asset i lowers its covariances
  - this shrinks the large estimates
  - why there: the unconstrained GMV shorts assets with high covariances with the others, so $w_i \geq 0$ binds on exactly those assets, whose large covariance estimates are the most likely to be inflated by noise
- Imposing the "wrong" constraint (no short sales) can reduce out-of-sample risk
  - the effect is largest when covariances come from few observations

Constraints as Shrinkage: A Two-Asset Example

- $\sigma_1 = 25\%$, $\sigma_2 = 10\%$, $\rho = 0.5$: $s_{11} = 0.0625$, $s_{22} = 0.0100$, $s_{12} = 0.0125$
- Unconstrained GMV: $w_1^* = (s_{22} - s_{12})/(s_{11} + s_{22} - 2s_{12}) = -0.0025/0.0475 = -0.053$
  - the volatile asset is shorted because it co-moves strongly with the other; volatility 9.93%
- Long-only: $w_1 \geq 0$ binds, w $= (0, 1)$, volatility 10%
  - first-order condition: Sw $= (s_{12}, s_{22}) = (0.0125, 0.0100) = \lambda_0 1 + \boldsymbol{\lambda}$, so $\lambda_0 = 0.0100$, $\boldsymbol{\lambda} = (0.0025, 0)$
- Modified matrix: $\tilde{s}_{12} = 0.0125 - 0.0025 = 0.0100$
  - $\tilde{s}_{11} = 0.0625 - 2(0.0025) = 0.0575$, $\tilde{s}_{22} = 0.0100$
  - correlation lowered from 0.5 to $0.0100/\sqrt{0.0575 \times 0.0100} \approx 0.42$; the variance of asset 1 also falls
  - unconstrained GMV of $\tilde{\mathrm{S}}$: $w_1^* = (0.0100 - 0.0100)/(\cdots) = 0$, the long-only solution

Nonlinear Shrinkage

- Linear shrinkage mixes two matrices with one δ
  - **nonlinear** shrinkage keeps the sample eigenvectors and corrects each eigenvalue separately
  - $\hat{\Sigma}_{\text{NL}} = \text{U}\,\text{diag}(\tilde{d}_1, \ldots, \tilde{d}_N)\text{U}^\top$, U: the eigenvectors of S, $\tilde{d}_i$: the corrected eigenvalues
  - Ledoit & Wolf (2017): tuned for portfolio selection; Ledoit & Wolf (2020): an analytical formula, no optimisation
- Out of sample, 60-month window, GMV volatility
  - Sectors: sample 11.90%, linear (constant correlation) 11.80%, nonlinear 11.62%
  - 16 ETFs: sample 4.06%, linear 4.38%, nonlinear 4.31%
- A constant correlation fits neither stock–stock (positive) nor stock–bond (negative) pairs
  - the target matters (Seminar 4, B7)
- No estimator dominates
  - nonlinear wins on sectors, the sample matrix on 16 ETFs
  - the choice is made out of sample
- At $N = 16$, $T = 60$ the sample matrix is still competitive
  - it becomes unusable as N approaches T

From Covariance Shrinkage to Mean Shrinkage



Fischer Black (1938–1995)

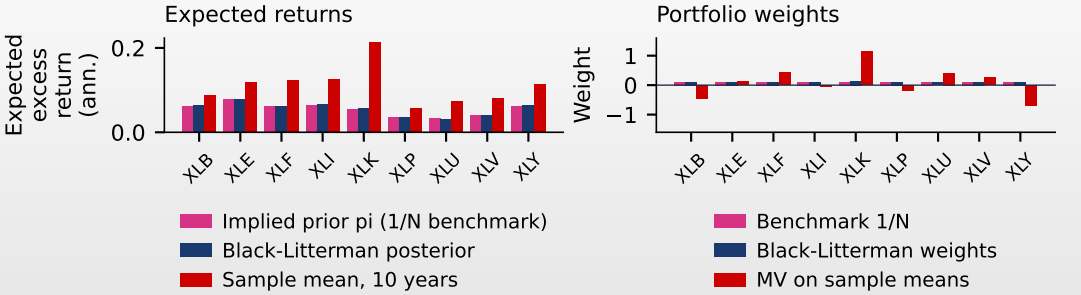
Photo: Dalmatine at English Wikipedia; public domain; Wikimedia Commons

- The question of this section: which means do we use, if the sample means are too imprecise?
- Black & Litterman (1992) shrinks the means towards those implied by a **benchmark** portfolio w_b
- Reverse optimisation: a mean–variance investor with risk aversion $\delta_{\mathbf{RA}}$ and access to cash
  - holds $w^* = (\delta_{\mathbf{RA}}\boldsymbol{\Sigma})^{-1}\boldsymbol{\mu}$ in risky assets
  - the returns that make w_b optimal are

$$\boldsymbol{\pi} = \delta_{\mathbf{RA}}\,\boldsymbol{\Sigma}\,w_b$$

- $\boldsymbol{\pi}$: **implied** (equilibrium) excess returns
  - with market weights, the CAPM returns
- Here: no market capitalisations for the sectors, so $w_b = 1/N$, $\delta_{\mathbf{RA}} = 2.5$
- Then add the investor's **views**, weighted by their confidence

Black–Litterman on US Sectors (1/2)



MFM_ch4_black_litterman

Black–Litterman on US Sectors (2/2)

- Left: annualised means
  - the implied means $\boldsymbol{\pi}$
  - the Black–Litterman means after the view
  - the 10-year sample means
- Right: the 1/N, Black–Litterman and sample-mean MV weights
- $\boldsymbol{\Sigma}$ from monthly excess returns, August 2016 – July 2026
  - 120 months, the other window of DeMiguel, Garlappi & Uppal (2009)
- One relative view: XLK minus XLU $= 3\%$ a year (view portfolio: $+1$ XLK, -1 XLU)
- XLK minus XLU: 2.2% implied, 2.6% after combining with the view (formula on the next slide)
- Weights: XLK from 11.1% to 14.4%, XLU from 11.1% to 7.8%
  - all others unchanged (by hand: Seminar 4, A7)
- With sample means instead (XLK: 21.3% a year)
  - the MV weights become $+115\%$ XLK, -68% XLY and -45% XLB

MFM_ch4_black_litterman

Views and the Posterior

- Prior: $\boldsymbol{\mu} \sim N(\boldsymbol{\pi}, \tau\boldsymbol{\Sigma})$
  - τ scales the uncertainty about $\boldsymbol{\pi}$
- K views: $\mathrm{P}\boldsymbol{\mu} = \mathrm{q} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N(0, \boldsymbol{\Omega})$
  - P: $K \times N$ matrix, row k = the portfolio of view k
  - q: the view returns; $\boldsymbol{\Omega}$: their uncertainty
- Posterior mean, a precision-weighted average

$$\boldsymbol{\mu}_{\mathbf{BL}} = \left[(\tau\boldsymbol{\Sigma})^{-1} + \mathrm{P}^{\top}\boldsymbol{\Omega}^{-1}\mathrm{P}\right]^{-1}\left[(\tau\boldsymbol{\Sigma})^{-1}\boldsymbol{\pi} + \mathrm{P}^{\top}\boldsymbol{\Omega}^{-1}\mathrm{q}\right]$$

- Black–Litterman weights: $\mathrm{w}_{\mathbf{BL}} = (\delta_{\mathbf{RA}}\boldsymbol{\Sigma})^{-1}\boldsymbol{\mu}_{\mathbf{BL}}$
- One asset, $\pi = 5\%$, $\tau\sigma^{2} = 0.00128$, view 8% with variance 0.0004
  - view weight $0.00128/(0.00128 + 0.0004) = 0.76$, so $\mu_{\mathbf{BL}} = 7.29\%$

Black–Litterman: Limiting Cases

- $\boldsymbol{\Omega} \to 0$: the views hold exactly, $\mathrm{P}\boldsymbol{\mu}_{\mathrm{BL}} = \mathrm{q}$
- $\boldsymbol{\Omega} \to \infty$: the views are ignored, $\boldsymbol{\mu}_{\mathrm{BL}} = \boldsymbol{\pi}$
- Without views, the benchmark w_b itself
- With the same $\boldsymbol{\Sigma}$ in the weights, only the assets in the views move
- The same estimator is a mixed regression estimated by generalised least squares (GLS) (Theil & Goldberger (1961))
  - derivation: Appendix; Seminar 4, A6

Black–Litterman: What δ_{RA} and τ Do

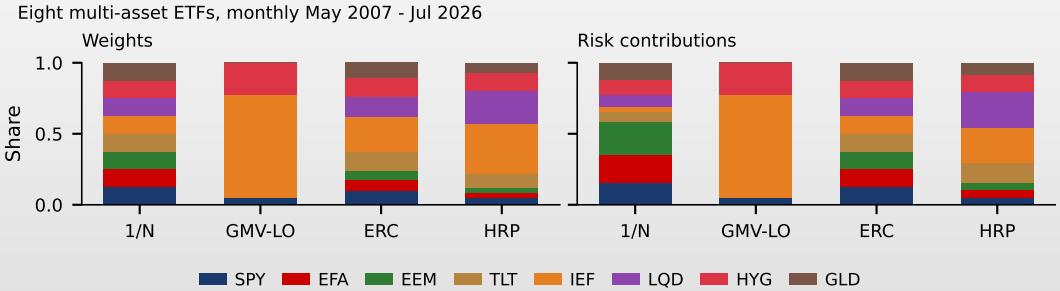
- The values of He & Litterman (2002): $\delta_{\text{RA}} = 2.5$, $\tau = 0.05$, $\boldsymbol{\Omega} = \text{diag}(\text{P}\,\tau\boldsymbol{\Sigma}\,\text{P}^\top)$
- **Risk aversion**: premultiply $\boldsymbol{\pi} = \delta_{\text{RA}}\boldsymbol{\Sigma}\text{w}_b$ by $\text{w}_b^\top$

$$\delta_{\text{RA}} = \frac{\text{w}_b^\top \boldsymbol{\pi}}{\text{w}_b^\top \boldsymbol{\Sigma}\text{w}_b} = \frac{\text{benchmark excess return}}{\text{benchmark variance}}$$

  - e.g. 6% excess return at 15.5% volatility: $0.06/0.155^2 = 2.5$
  - $\boldsymbol{\pi}$ is linear in δ_{RA}: doubling it doubles the implied XLK minus XLU spread of 2.2%
- **Prior uncertainty**: $\tau\boldsymbol{\Sigma}$ is the uncertainty about the mean
  - one reading is $\tau \approx 1/Y$, the variance of a mean estimated over Y years relative to $\boldsymbol{\Sigma}$ (Merton's argument)
  - so $\tau = 0.05$ is about 20 years of data
- With this $\boldsymbol{\Omega}$ and one view, τ cancels
  - the posterior spread is the midpoint $(2.2\% + 3\%)/2 = 2.6\%$
  - the weights do not depend on τ
- The confidence in the view is set by $\boldsymbol{\Omega}$ relative to $\tau\boldsymbol{\Sigma}$, not by τ alone

Weights versus Risk Contributions (1/2)

The question of this section: can we build weights from Σ alone, without means and without inverting Σ?



MFM_ch4_risk_contrib

Weights versus Risk Contributions (2/2)

- Eight multi-asset ETFs, full-sample covariance, May 2007 – July 2026 ($T = 231$)
  - volatilities from 6.7% (IEF) to 21.1% (EEM)
- Left: the weights; right: each asset's share of risk
  - share of risk: $w_i(\Sigma \mathrm{w})_i / \sigma_p^2$ (next slide)
  - shares in decimals ($0.5 = 50\%$)
- 1/N: equities and high yield receive half of the weight but 69% of the risk
- ERC (equal risk contribution): 12.5% of the risk from every asset
  - IEF gets the largest weight (25%)
- Long-only GMV: 73% IEF, 22% HYG, 5% SPY, five assets at zero
- HRP (Hierarchical Risk Parity): largest weights IEF (35%) and LQD (23%)
- Volatility: 1/N 9.4%, ERC 7.9%, HRP 7.1%, long-only GMV 5.8%

MFM_ch4_risk_contrib

Risk Contributions

- Volatility is homogeneous of degree one in the weights (Euler)

$$\sigma_p = \sum_{i=1}^{N} w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^{N} \frac{w_i (\Sigma \mathrm{w})_i}{\sigma_p}$$

- **Risk contribution** of asset i: $RC_i = w_i(\Sigma \mathrm{w})_i / \sigma_p$
  - percentage share $RC_i/\sigma_p = w_i(\Sigma \mathrm{w})_i/\sigma_p^2$
- Worked example, 50/50 in SPY and IEF (monthly 2002–2026, annualised)
  - $\sigma_1^2 = 0.02166$, $\sigma_2^2 = 0.00441$, $\sigma_{12} = -0.00098$
  - $(\Sigma \mathrm{w})_1 = 0.5(0.02166) + 0.5(-0.00098) = 0.01034$, $(\Sigma \mathrm{w})_2 = 0.5(-0.00098) + 0.5(0.00441) = 0.00172$
  - $\sigma_p^2 = 0.5(0.01034) + 0.5(0.00172) = 0.00603$, $\sigma_p = 7.76\%$; shares: SPY 85.8%, IEF 14.2%
- Risk budgeting chooses w so that the shares equal preset budgets b_i

Equal Risk Contribution

- ERC (Maillard, Roncalli & Teïletche (2010)): $RC_i = RC_j$ for all i, j
  - computed from a convex problem, then normalised

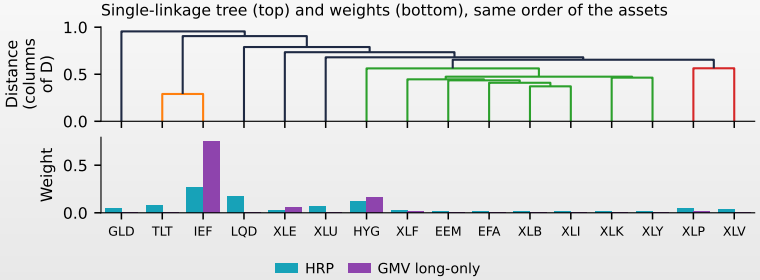
$$\min_{\mathrm{y}>0} \ \tfrac{1}{2}\mathrm{y}^\top \boldsymbol{\Sigma}\mathrm{y} - \tfrac{1}{N}\textstyle\sum_i \ln y_i, \qquad \mathrm{w} = \mathrm{y}/1^\top\mathrm{y}$$

  - y: unnormalised positive weights; the log term keeps every $y_i > 0$ and its first-order condition gives $y_i(\boldsymbol{\Sigma}\mathrm{y})_i = 1/N$ for all i
- For $\boldsymbol{\Sigma}$ positive definite a solution exists and is unique
  - strictly convex objective, infinite at the boundary
- Volatility ordering: $\sigma_{\mathrm{GMV}} \leq \sigma_{\mathrm{ERC}} \leq \sigma_{1/N}$
- Two assets: ERC is **inverse volatility**, $w_1 = \sigma_1^{-1}/(\sigma_1^{-1} + \sigma_2^{-1})$
  - for any $\rho \in (-1, 1)$
  - SPY (14.7%) and IEF (6.6%): ERC holds 31% SPY, volatility 6.1%, each asset 50% of the variance
- ERC is the tangency portfolio if all assets have the same Sharpe ratio and equal correlations

Hierarchical Risk Parity: From a Tree to Weights

- HRP (López de Prado (2016)): three steps, no matrix inversion, all weights positive
  - works even when $\boldsymbol{\Sigma}$ is singular
- **Step 1, tree**
  - correlation distance $d_{ij} = \sqrt{(1-\rho_{ij})/2} \in [0,1]$; ρ_{ij}: correlation of assets i and j
  - Euclidean distance between the **columns** of $\mathrm{D} = (d_{ij})$: $\tilde{d}_{ij} = \sqrt{\sum_k (d_{ki} - d_{kj})^2}$
  - single linkage on $\tilde{d}_{ij}$ (repeatedly merge the two closest clusters) gives a **dendrogram**
- **Step 2, quasi-diagonalisation**
  - order the assets as the leaves of the tree, so that similar ones are adjacent
- **Step 3, recursive bisection**: split the **ordered list** in two halves
  - the halves get α and $1-\alpha$, $\alpha = 1 - V_1/(V_1+V_2)$, $V_j = \tilde{\mathrm{w}}_j^\top \boldsymbol{\Sigma}_j \tilde{\mathrm{w}}_j$ the variance of half j
  - $\tilde{\mathrm{w}}_j$: weights $\tilde{w}_i \propto 1/\sigma_i^2$ inside half j, $\boldsymbol{\Sigma}_j$ its covariance
  - repeat inside each half until single assets remain
- Mantegna (1999): the same distance and the **minimum spanning tree** reveal the hierarchy of a stock market
- Textbook: Härdle & Simar (2019), cluster analysis
- A worked four-asset example: Seminar 4, B4

HRP on 16 ETFs



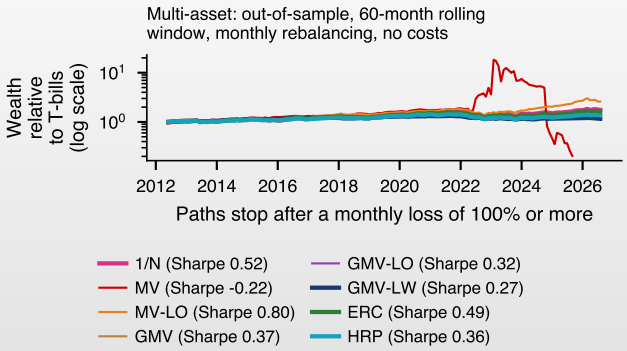
- Nine sectors and seven multi-asset ETFs, monthly, May 2007 – July 2026
  - closest pair TLT and IEF ($\tilde{d} = 0.29$), gold joins last (0.96)
- Equity cluster: cyclical sectors, EFA, EEM and high-yield credit
  - defensive sectors (utilities, staples, health care) and energy join it late
- HRP: all 16 weights positive, largest IEF 26.7%, LQD 17.6%, HYG 12.6%
  - long-only GMV: 75.6% in IEF and only five assets above zero

MFM_ch4_hrp

Backtest Design

- The question of this section: does an estimated rule beat 1/N
  - when the weights use only the data available at the decision date?
- The weights of month t use months $t-60, \ldots, t-1$ only (the 60-month rolling window)
- Monthly rebalancing, at the data frequency
  - its cost is measured separately, through turnover
- Eight rules: 1/N, MV, MV-LO, GMV, GMV-LO, GMV-LW, ERC, HRP
- Three universes
  - Sectors: OOS January 2004 – July 2026, 271 months (first window January 1999 – December 2003)
  - Multi-asset and combined: OOS May 2012 – July 2026, 171 months
- Measures
  - annualised Sharpe $\sqrt{12}\,\bar{r}^e/\hat{\sigma}(r^e)$ from excess returns
  - maximum drawdown from total returns
  - every difference from 1/N with a test, corrected for multiple testing

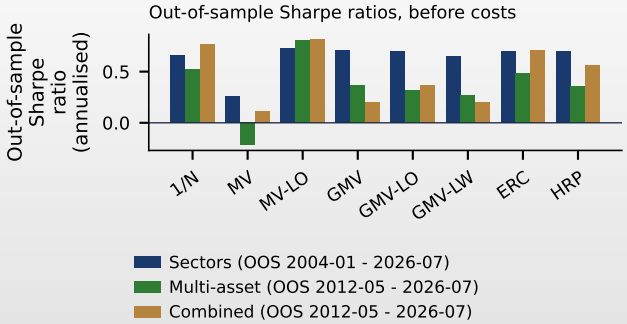
Out-of-Sample Growth: Multi-Asset



- Vertical axis: wealth relative to a Treasury-bill account, log scale
  - eight multi-asset ETFs, before costs
- MV collapses: volatility 518% a year, turnover 44 times wealth per month
  - a monthly loss above 100% ends its wealth path
- MV-LO has the highest Sharpe (0.80); 1/N 0.52; ERC 0.49
- GMV rules: lowest volatility (3.8%–5.4%), low Sharpe (0.27–0.37)
  - they load on Treasuries, whose excess returns were slightly negative after May 2012 (TLT −0.5%, IEF −0.2% a year)
- Is MV-LO really better, or lucky? A test is needed

MFM_ch4_oos_backtest

Sharpe Ratios Across Universes



- Annualised, before costs
  - complete tables for all three universes in the ▸ Appendix
- Sectors: every rule except MV lands within 0.07 of 1/N (0.66)
  - MV 0.26 with drawdown −100%
- 1/N has 0.66 here (2004–2026), not 0.54 as in the simulations (1999–2026)
  - the periods differ
- Combined: 1/N 0.76, MV-LO 0.81, ERC 0.71
  - the GMV rules 0.20–0.36
- Turnover per month, sectors: 1/N 0.02, ERC 0.03, GMV-LW 0.14, MV-LO 0.23, MV 3.86

MFM_ch4_oos_backtest

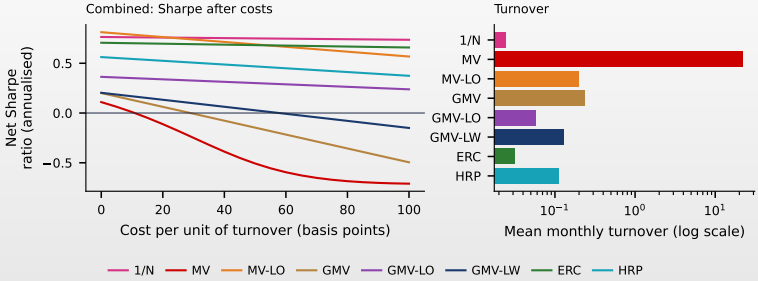
Turnover and Costs: One Worked Month (1/2)

- Between rebalancings weights drift with returns: $w_{i,t}^{+} = w_{i,t}(1 + R_{i,t}) / \sum_j w_{j,t}(1 + R_{j,t})$
  - $R_{i,t}$: **total** return of asset i in month t; the denominator is $1 + R_{p,t}$, one plus the portfolio's total return
- **Turnover**: traded fraction of wealth, $TO_{t+1} = \sum_i |w_{i,t+1} - w_{i,t}^{+}|$
  - summed over all N assets
- Worked month, sectors, 1/N: in January 2004 XLB returned -4.4% and XLK $+4.3\%$
  - so the weights of 11.11% drifted to 10.50% and 11.45%
  - 1/N total return in January 2004: 1.18% (excess 1.11% plus the Treasury bill 0.07%; Seminar 4, B1)
  - XLB: $\frac{1}{9}(1 - 0.044)/1.0118 = 10.50\%$; XLK: $\frac{1}{9}(1 + 0.043)/1.0118 = 11.45\%$

Turnover and Costs: One Worked Month (2/2)

- Rebalancing back to 1/N in February 2004 trades $TO = 0.019$
  - 1/N also trades, selling winners and buying losers
  - XLB and XLK contribute $|0.1111 - 0.1050| + |0.1111 - 0.1145| = 0.0061 + 0.0034 = 0.0095$; the other seven sectors about 0.009 together
  - MV-LO, re-estimated on the new window, trades 0.748 in the same month
- Net return with cost c per unit traded: $R^{\text{net}}_{p,t} = R_{p,t} - c\, TO_t$
  - 1 basis point (bp) $= 0.01\%$, so 20 bp means $c = 0.002$
  - At 20 bp the February trade costs MV-LO $0.002 \times 0.748 = 0.15\%$ of wealth, 1/N $0.002 \times 0.019 = 0.004\%$

Sharpe Ratios after Costs



- Combined universe (16 ETFs)
  - left: net Sharpe (vertical) against the cost c, 0–100 bp per unit of turnover (horizontal)
  - right: mean monthly turnover, log scale, from 0.02 (1/N) to 22 (MV)
- Net Sharpe turns negative at about 11 bp for MV, 29 bp for GMV, 58 bp for GMV-LW
  - MV-LO overtakes 1/N only for costs below about 22 bp (Seminar 4, B2)
- 1/N and ERC barely move: low turnover is an advantage

MFM_ch4_costs

Inference on One Sharpe Ratio: Lo (2002)

- Lo (2002), i.i.d. Normal returns: $\sqrt{T}(\widehat{SR} - SR) \to N(0,\ 1 + SR^2/2)$
  - SR monthly, T the number of months
  - the standard error is $\sqrt{(1 + SR^2/2)/T}$
- 1/N on sectors, $T = 271$: monthly $\widehat{SR} = 0.191$, standard error 0.061
  - Annualised $0.66 \pm 1.96 \times 0.21$: 95% interval $[0.25, 1.08]$
- Annualising by $\sqrt{q}$ ($q = 12$ months per year) is exact only without autocorrelation
  - in general $SR(q) = \eta(q)\, SR$

$$\eta(q) = \frac{q}{\sqrt{q + 2\sum_{k=1}^{q-1}(q-k)\rho_k}}$$

  - ρ_k: autocorrelation of monthly returns at lag k; positive ρ_k lower η, negative ρ_k raise it
- For 1/N on sectors: $\rho_1 = -0.03$, weighted sum $\sum_{k=1}^{11}(12-k)\rho_k = -0.50$
  - $\eta = 12/\sqrt{12 - 2(0.50)} = 12/\sqrt{11.0} = 3.62$ instead of $\sqrt{12} = 3.46$, annual Sharpe $0.191 \times 3.62 = 0.69$

Testing a Difference in Sharpe Ratios

- H_0: $SR_1 = SR_2$ for two strategies evaluated on the same months
- Jobson & Korkie (1981), in the form corrected by Memmel (2003) and given in Ledoit & Wolf (2008)
  - for i.i.d. Normal returns: $z = \hat{\Delta}/\sqrt{\hat{\theta}/T}$
  - $\hat{\theta} = 2(1-\hat{\rho}) + \frac{1}{2}\left(\widehat{SR}_1^2 + \widehat{SR}_2^2 - 2\widehat{SR}_1\widehat{SR}_2\hat{\rho}^2\right)$
  - $\hat{\rho}$: the correlation of the two strategies
- Ledoit & Wolf (2008): a robust version, by the delta method

$$\hat{\Delta} = \widehat{SR}_1 - \widehat{SR}_2 = f(\hat{\mathrm{v}}), \qquad f(\mathrm{v}) = \frac{\mu_1}{\sqrt{\mathbb{E}r_1^2 - \mu_1^2}} - \frac{\mu_2}{\sqrt{\mathbb{E}r_2^2 - \mu_2^2}}$$

  - r_{1t}, r_{2t}: excess returns of the two strategies; μ_1, μ_2: their means; $\mathrm{v} = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$
  - standard error $\sqrt{\nabla f^\top \hat{\mathbf{\Psi}} \nabla f / T}$
  - $\hat{\mathbf{\Psi}}$: a HAC (heteroskedasticity and autocorrelation consistent) covariance of the four moments
- Strategies on the same assets are highly correlated: small differences can be significant

The HAC Test Step by Step

- Gradient of f with respect to $\mathrm{v} = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$
  - writing $\sigma_i^2 = \mathbb{E}[r_i^2] - \mu_i^2$

$$\nabla f = \Big(\frac{\mathbb{E}[r_1^2]}{\sigma_1^3},\ -\frac{\mathbb{E}[r_2^2]}{\sigma_2^3},\ -\frac{\mu_1}{2\sigma_1^3},\ \frac{\mu_2}{2\sigma_2^3} \Big)$$

  - these are derivatives in the **uncentred** second moments; a gradient in (μ_i, σ_i^2) does not fit this $\hat{\mathbf{\Psi}}$
- $\hat{\mathbf{\Psi}}$: long-run covariance of $\mathrm{y}_t = (r_{1t} - \hat{\mu}_1,\ r_{2t} - \hat{\mu}_2,\ r_{1t}^2 - \widehat{\mathbb{E}r_1^2},\ r_{2t}^2 - \widehat{\mathbb{E}r_2^2})$
- HAC p-values in this chapter's tables: Bartlett (Newey–West) kernel
  - $\lfloor 4(T/100)^{2/9} \rfloor = 4$ lags for every T from 107 to 271
  - $z = \hat{\Delta}/\mathrm{SE}$, $p = 2[1 - \Phi(|z|)]$
- The bootstrap studentizes with a different SE (prewhitened quadratic-spectral kernel, Appendix)
  - so the two statistics differ slightly
  - sectors, MV-LO minus 1/N: $\hat{\Delta} = 0.063$; Bartlett SE 0.120, $z = 0.53$, HAC $p = 0.60$; quadratic-spectral (QS) SE 0.127, studentized statistic 0.50

The Block Bootstrap

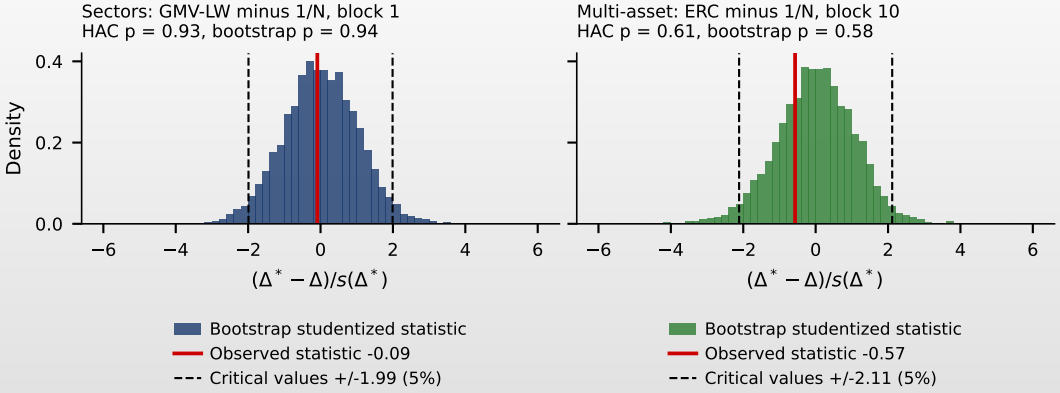


Bradley Efron, inventor of the bootstrap (1979)

Photo: Ryan K. Morris (2007); public domain; Wikimedia Commons

- Returns are dependent (volatility clustering)
  - resample **blocks** of consecutive months (Künsch (1989))
- Ledoit & Wolf (2008), Section 3.2.2: circular blocks of **pairs** (r_{1t}, r_{2t})
  - wrapping around the end of the sample, so the correlation of the two strategies is kept
- The statistic in draw m: $\tilde{d}^{*m} = |\hat{\Delta}^{*m} - \hat{\Delta}| / s(\hat{\Delta}^{*m})$
  - centred at $\hat{\Delta}$ and divided by the standard error $s(\cdot)$
- Asymptotically pivotal (its limit distribution has no unknown parameters)
  - more accurate than the Normal approximation of the HAC test
- $p = \left[\#\{m : \tilde{d}^{*m} \geq |\hat{\Delta}| / s(\hat{\Delta})\} + 1\right] / (M + 1)$, with $M = 4999$ draws
- Block length chosen by the calibration of Ledoit & Wolf (2008), Algorithm 3.1 (▸ Appendix)
  - 1 for six of the seven sector rules
  - between 1 and 10, most often 10, for multi-asset and combined rules

Is Anything Better than 1/N? (1/2)



MFM_ch4_sharpe_test

Is Anything Better than 1/N? (2/2)

- Horizontal axis: $(\Delta^* - \hat{\Delta})/s(\Delta^*)$
  - the bootstrap distribution of the studentized difference against 1/N
- Red: the observed statistic $\hat{\Delta}/s(\hat{\Delta})$
  - -0.09 (sectors, GMV-LW) and -0.57 (multi-asset, ERC)
  - dashed: bootstrap critical values ± 1.99 and ± 2.11
- Sectors, GMV-LW minus 1/N: -0.02, 95% interval $[-0.36, 0.33]$
  - HAC $p = 0.93$, bootstrap $p = 0.94$ (block 1)
- Multi-asset, ERC minus 1/N: -0.03, $[-0.15, 0.08]$
  - HAC $p = 0.61$, bootstrap $p = 0.58$ (block 10)
  - so correlated with 1/N that the interval is narrow, yet it covers zero
- Sectors: no rule differs from 1/N at 5%
  - the smallest p-value is ERC's ($+0.04$, HAC $p = 0.09$)
- Multi-asset: MV-LO $+0.29$ is **not** significant (HAC $p = 0.12$, bootstrap 0.16)

MFM_ch4_sharpe_test

Many Rules, Many Tests (1/2)

Universe	Rule minus 1/N	$\hat{\Delta}$	HAC p	Bootstrap p
Multi-asset	MV	-0.74	0.0138	0.0566
Multi-asset	HRP	-0.16	0.0720	0.0460
Combined	GMV	-0.56	0.0304	0.0496
Combined	GMV-LO	-0.40	0.0477	0.0804
Combined	GMV-LW	-0.56	0.0238	0.0322
Combined	MV-LO	$+0.05$	0.8175	0.8180

- No rule beats 1/N significantly in any universe
- Without a multiple-testing adjustment, at 5%
  - four rules have a significantly lower Sharpe than 1/N by the HAC test (multi-asset MV; combined GMV, GMV-LO, GMV-LW)
  - three by the bootstrap (multi-asset HRP; combined GMV, GMV-LW)
- The GMV rules have low risk, but even lower return
- The table shows 6 of the 21 two-sided tests, H_0: $\Delta_k = 0$
  - 7 rules $\times$ 3 universes; all 21 in the Appendix

Many Rules, Many Tests (2/2)

- Holm over all 21: smallest adjusted p is 0.29 (HAC, 21×0.014) and 0.68 (bootstrap, 21×0.032)
  - nothing survives (next slide; Chapter 3)
- A different question, the best of the 24 configurations (8 rules $\times$ 3 universes)
  - the deflated Sharpe ratio (DSR) of Bailey & López de Prado (2014) gives $0.84 < 0.95$ for MV-LO on 16 ETFs
  - but the verdict hinges on the effective number of trials (▸ Appendix)

Holm and Benjamini–Hochberg on This Chapter's Tests

- **FWER** (family-wise error rate): the probability of at least one false rejection in the family
- **FDR** (false discovery rate): the expected share of false rejections among the rejections
- Sort the m p-values, $p_{(1)} \leq \cdots \leq p_{(m)}$
  - **Holm** (controls the FWER under any dependence): $\tilde{p}_{(i)} = \max_{j \leq i} \min\{1, (m - j + 1)\, p_{(j)}\}$
  - **Benjamini–Hochberg** (Benjamini & Hochberg (1995); controls the FDR): $\tilde{p}_{(i)} = \min_{j \geq i} \min\{1, \frac{m}{j}\, p_{(j)}\}$
  - reject where $\tilde{p} < 0.05$ (Holm) or $\tilde{p} <$ the chosen FDR (BH)
- The four smallest two-sided HAC p-values, $m = 21$

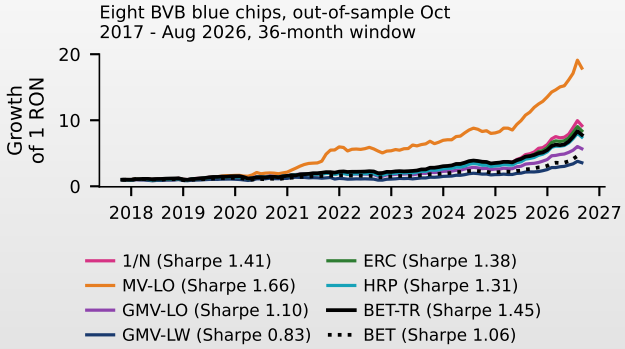
Test	$p_{(j)}$	**Holm**	$\frac{21}{j}\, p_{(j)}$
Multi-asset, MV	0.0138	$21 \times 0.0138 = 0.289$	0.289
Combined, GMV-LW	0.0238	$20 \times 0.0238 = 0.475$	0.249
Combined, GMV	0.0304	$19 \times 0.0304 = 0.577$	0.213
Combined, GMV-LO	0.0477	$18 \times 0.0477 = 0.858$	0.250

- BH takes the minimum from the bottom up: the first three all become 0.213
  - no rejection at an FDR of 10%

Setup: a Romanian Investor

- The question of this section: does an allocation rule beat the BET-TR index on BVB blue chips
  - for an investor in RON?
- Eight blue chips with complete histories since October 2014: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL
- A 36-month window, not 60: with 60 months only 83 months would remain out of sample
  - first portfolio October 2017
- Rules: 1/N, MV-LO, GMV-LO, GMV-LW, ERC, HRP
  - no unconstrained MV and GMV, whose weights would be extreme on 36-month windows
- Benchmark: BET-TR (total return, with dividends)
  - BET is shown only to measure the dividend effect
- Sharpe ratios with $R_f = 0$ (no RON money-market series in the data)
  - hence higher than excess-return Sharpe ratios
- Earlier Romanian evidence on diversification: Dragotă, Șerbănescu & Pele (2008)

BVB Portfolios versus BET-TR (1/2)



- Chart: the value of RON 1 invested in October 2017
  - out of sample to August 2026 (107 months), before costs
- BET-TR: mean 24.7% a year, volatility 17.0%, Sharpe 1.45
- BET (no dividends): mean 17.8%, Sharpe 1.06
- Close to BET-TR: 1/N 1.41, ERC 1.38, HRP 1.31

MFM_ch4_bvb

BVB Portfolios versus BET-TR (2/2)

- MV-LO: Sharpe 1.66, $+0.21$ over BET-TR
  - HAC $p = 0.44$, bootstrap 0.53
- GMV-LW: Sharpe 0.83, -0.63 below BET-TR
  - HAC $p = 0.050$, bootstrap 0.10
- Average correlation of the eight stocks 0.44 limits diversification
  - the latest GMV-LO puts 70% in TLV (Seminar 4, C2)

MFM_ch4_bvb

Which Frontier Can a Large Investor Reach? (1/2)

- The question: which allocation rule stays efficient when trading costs grow with the size of the investor?
- Jensen, Kelly, Malamud & Pedersen (2026), *Review of Financial Studies*
  - how should portfolios be built from machine-learning (ML) return forecasts when every trade moves prices?
- **Implementable efficient frontier** (eq. 12)
  - for each volatility σ, the highest expected return **net of trading costs**

$$k(\sigma) = \max_{\pi} \; E[r_{t+1}^{\text{net}}] \quad \text{s.t.} \quad E[\pi_t^\top \boldsymbol{\Sigma} \pi_t] = \sigma^2$$

  - π_t: vector of risky-asset weights at t (the paper's notation; not the implied returns $\boldsymbol{\pi}$ of Black–Litterman); r_{t+1}^{net}: portfolio return after trading costs
  - $k(\sigma)$: the best net expected return at volatility σ
- Proposition 1: the net Sharpe ratio $k(\sigma)/\sigma$ falls along the frontier
  - a larger investor faces a lower frontier

Which Frontier Can a Large Investor Reach? (2/2)

- Cost as a fraction of wealth W_t (eqs. 4, 7), quadratic in the trade

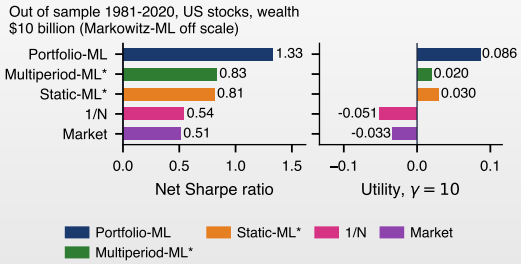
$$c_t = \tfrac{W_t}{2}(\pi_t - g_t\pi_{t-\mathbf{1}})^\top \mathbf{\Lambda}(\pi_t - g_t\pi_{t-\mathbf{1}}), \qquad \mathbf{\Lambda} = \mathrm{diag}(0.2/V_i)$$

  - g_t: drifts last month's weights; V_i: the average daily dollar volume of asset i
- Optimal rule: trade only part of the way towards an **aim portfolio** A_t (Gârleanu & Pedersen (2013))

$$\pi_t = m\, g_t\pi_{t-\mathbf{1}} + (\mathbf{I} - m)A_t$$

  - A_t: the portfolio one would hold if trading were free, a weighted average of today's and expected future Markowitz portfolios
  - m: an $N \times N$ matrix of trade rates (eq. 14), so $\mathbf{I} - m$ is a matrix too; in the scalar special case of Gârleanu & Pedersen (2013), m is the share of the old position kept each period
  - Portfolio-ML learns A_t from the signals by a ridge regression (least squares with a penalty on the size of the coefficients) whose objective is utility net of costs

The Paper's Results: Net Sharpe and Utility



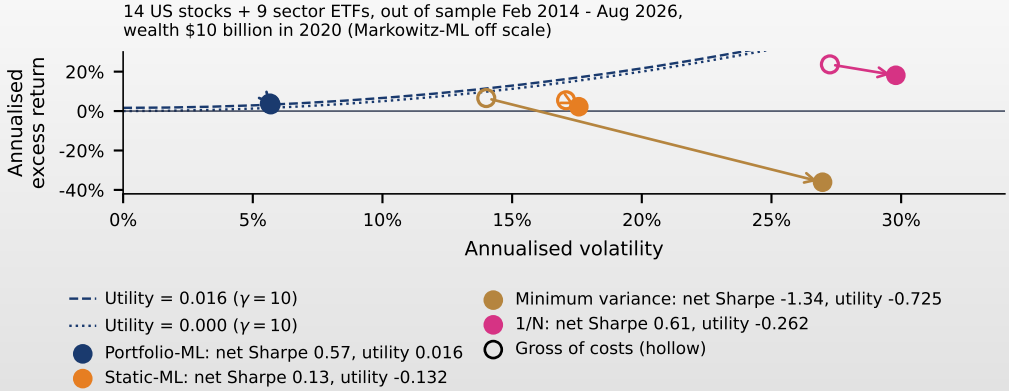
- Jensen, Kelly, Malamud & Pedersen (2026), Table 2: net of costs, 1981–2020, $\gamma = 10$
  - utility (eq. 11) = annual net return $-\frac{\gamma}{2}\times$ annual variance
- Markowitz-ML (mean–variance on ML return forecasts, no costs)
  - gross Sharpe 2.06, turnover 79.36, leverage 74.61: net return extremely negative
- Portfolio-ML: net Sharpe 1.33, utility 0.086 (net return 15%, volatility 11%, cost 0.6% a year)
- Multiperiod-ML* (forecasts for 12 horizons): net Sharpe 0.83, utility 0.020
- Static-ML* (one-period costs; * = extra tuning layer) 0.81, 0.030
  - 1/N 0.54, −0.051; market 0.51, −0.033
  - only cost-aware rules have positive utility

MFM_ch4_implementable_frontier

Paper and Course Application: What Is Compared?

	Paper	**Course replication**
Assets	US stocks with market capitalisation above the NYSE (New York Stock Exchange) median (184–805)	14 US stocks + 9 sector ETFs
Signals	115 characteristics (Jensen, Kelly & Pedersen (2023))	5 price and volume signals
Out of sample	1981–2020, monthly	February 2014 – August 2026 (151 months)
Investor	$\gamma = 10$, wealth \$10 billion at the end of 2020, growing with the market	same; wealth scaled by the cumulative SPY return before and after 2020
Utility	annual net return $-\frac{\gamma}{2}\times$ annual variance	same: $12\times$ monthly mean $-\frac{\gamma}{2}\times 12\times$ monthly variance

Gross and Net of Costs on the Course Data (1/2)



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Gross and Net of Costs on the Course Data (2/2)

- Axes: annualised volatility (horizontal) and annualised excess return (vertical)
  - hollow marker: before costs
  - filled: after the cost c_t (eq. 35 of the paper)
- Dashed curve: the utility of Portfolio-ML (0.016)
  - dotted curve: utility 0
- Portfolio-ML: return 3.8% gross and 3.3% net, volatility 5.7%, net Sharpe 0.57
- Utility of Portfolio-ML: $0.0326 - \frac{10}{2} \times 0.0569^2 = 0.016$ (net mean and net volatility, annual)
- 1/N: 18.1% net and Sharpe 0.61, but its 29.8% volatility gives a utility of -0.26
- Minimum variance: $+6.6\%$ gross, -36.1% net
  - its leverage of 2.6 costs 43% a year
- Limits: 23 assets, 5 signals, no factor model for $\boldsymbol{\Sigma}$ (Seminar 4, C3)

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Open Questions

- Questions
  - does any **estimated** allocation rule beat 1/N out of sample, net of costs?
  - does it still beat 1/N once every rule, window and universe tried is counted?
- Live evidence points both ways
  - Nonlinear shrinkage of Σ improves GMV out of sample in large cross-sections: Ledoit & Wolf (2017)
  - Combining tangency, GMV and 1/N under estimation risk: Kan, Wang & Zhou (2022)
  - Backtests need haircuts for the number of specifications tried: Harvey & Liu (2015)
- Formal statement: H_0: $\Delta_k \leq 0$ for all 21 (rule, universe) pairs
  - against H_1: $\Delta_k > 0$ for at least one k
  - with the FWER or FDR fixed for the whole family (Many Rules, Many Tests)
- One pass of the loop with an AI assistant
  - literature with DOIs, competing hypotheses, code on the course's monthly data
  - a pre-registered family of tests, a referee-style critique (prompts in the Appendix)
- Falsified by: a positive $\hat{\Delta}_k$ that survives a family-wise correction, costs and a second sample

What the Researcher Must Check

- **Look-ahead**: weights at t built with month-t returns
  - test: assert that the last estimation date is before t
- **Survivorship and selection**: an ETF list chosen today
  - test: use the universe available at each date, including later delisted ETFs, or disclose the limitation
- **Multiple testing**: 7 rules $\times$ 3 universes already give 21 tests
  - test: Holm (FWER) or Benjamini–Hochberg (FDR) adjusted p-values
- **Power**: for MV-LO minus 1/N on the multi-asset universe, $\hat{\Delta}$ has a HAC standard error of 0.18 on 171 months
  - test: compute the detectable effect before running
- **Return type**: portfolios aggregate **simple** returns, Sharpe ratios use **excess** returns
  - test: recompute one month step by step
- **Invented references**: an unresolvable DOI or a title mismatch
  - the claim is dropped and the error goes into `AI_ERRORS.md`

Mini-Case: Data Snooping in This Chapter's Backtest (1/2)

- A typical assistant proposal: "MV-LO beats 1/N on the multi-asset universe"
  - superiority is one-sided: H_0: $\Delta \leq 0$, $p = 1 - \Phi(\hat{\Delta}/\mathrm{SE})$
  - Φ the standard Normal distribution function, SE the standard error
- Replicate: $\hat{\Delta} = +0.29$, HAC standard error 0.18, one-sided $p = 0.06$
  - two-sided HAC 0.12, bootstrap 0.16
  - $z = 0.2863/0.1836 = 1.56$, $p = 1 - \Phi(1.56) = 0.06$
- Power 0.8 for $\Delta = 0.29$, one-sided: about 435 months (36 years) at 5%
  - 944 months (79 years) at the Holm level 5%/21
  - months needed: $T_{\mathrm{obs}}\left[(z_{1-\alpha} + z_{0.8})\,\mathrm{SE}/\Delta\right]^2 = 171\,\left[(1.645 + 0.842) \times 0.1836/0.2863\right]^2 \approx 435$
  - at $\alpha = 0.05/21$, $z_{1-\alpha} = 2.82$, which gives ≈ 944

MFM_ch4_ai_discovery

Mini-Case: Data Snooping in This Chapter's Backtest (2/2)

- Test the family, not the winner: 21 tests (7 rules $\times$ 3 universes)
  - one-sided p-values of all 21 in the Appendix
  - Only 7 of 21 differences are positive; the 4 two-sided HAC p-values below 0.05 are all **losses**, which never reject a superiority null
  - One-sided: smallest $p = 0.046$ (sectors, ERC); adjusted Holm 0.96, Benjamini–Hochberg 0.62; 0 discoveries at an FDR of 10%
- Why 0.96 here but 0.29 on Many Rules, Many Tests
  - a loss has a large one-sided p (about $1 - p_{\text{two-sided}}/2$, e.g. 0.99 for multi-asset MV)
  - so the significant losses drop out
  - the smallest one-sided p belongs to a gain, 0.046, and Holm multiplies it by 21
- Interpretation: the data cannot rank these rules
  - the bootstrap interval for MV-LO, $[-0.13, 0.70]$, admits losses and large gains
  - failure to reject is not proof of equality

MFM_ch4_ai_discovery

Your Turn: A Research Project Idea

- **Questions**
  - does nonlinear shrinkage (Ledoit & Wolf (2017)) beat 1/N on the 16 ETFs of this chapter, after 10–50 bp costs?
  - does a three-fund combination (Kan, Wang & Zhou (2022)) beat 1/N on the same ETFs, after the same costs?
- **Replicate first**
  - the combined-universe results (1/N Sharpe 0.76, MV-LO 0.81)
  - the adjusted p-values of the mini-case
- **Pre-register**, before any result
  - hypotheses, family of tests, error rate and power
  - a placebo: random long-only weights with the same turnover
- **Report**: code on the course's monthly data, a table of all specifications, a referee-style self-critique
- **Log**: prompts in `AI_USE.md`
  - every error caught and the test that caught it in `AI_ERRORS.md`

Conclusions: Which Rule, When?

Rule	Objective	Inputs	Main failure mode	Evidence in this chapter
MV	maximum Sharpe	$\hat{\boldsymbol{\mu}}$, $\hat{\boldsymbol{\Sigma}}^{-1}$	error maximisation, explosive weights	worst in every universe; insolvent path
MV-LO	maximum Sharpe, $w_i \geq 0$	$\hat{\boldsymbol{\mu}}$, $\hat{\boldsymbol{\Sigma}}$	corner solutions, turnover	best point estimates, never significant
GMV, GMV-LW	minimum variance	$\hat{\boldsymbol{\Sigma}}$ (shrunk)	loads on low-return assets	lowest risk, loses to 1/N on 16 ETFs (unadjusted)
ERC, HRP	balanced risk	$\hat{\boldsymbol{\Sigma}}$	a tilt of 1/N; HRP trades more	never significantly better than 1/N
Black–Litterman	benchmark plus views	w_b, views, $\boldsymbol{\Omega}$	confidence set too high	moves only the assets in the views
1/N	none	none	ignores risk differences	hard to beat after costs

- Evaluate every rule out of sample, net of costs
  - with a test of the Sharpe difference and a count of all tests
  - these verdicts are specific to these samples

Quick Checks with Toy Numbers (1/2)

- **Three rules, two assets**: $\sigma_1 = 30\%$, $\sigma_2 = 15\%$, $\rho = 0.2$
  - compute the weights and volatilities of GMV, ERC and 1/N; check $\sigma_{\text{GMV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}$
  - what share of the 1/N variance comes from asset 1? (Two Assets; Risk Contributions; Equal Risk Contribution)
- **Optimism of the plug-in Sharpe**: $N = 4$ assets, $T = 120$ months, true $\theta = 0.50$ a year
  - what in-sample maximum Sharpe (root mean square, annual) should you expect? (The Plug-in Sharpe Ratio Is Biased)
- **Noise band**: $N = 10$, $T = 40$, correlation matrix
  - compute the Marchenko–Pastur band; is a sample eigenvalue of 2.0 evidence of structure? (Why the Sample Covariance Matrix Fails)

Quick Checks with Toy Numbers (2/2)

- **One view, one asset**: $\pi = 4\%$, $\tau\sigma^2 = 0.0009$, view 10%
  - posterior mean and view weight for $\Omega = 0.0009$ and for $\Omega = 0.0003$ (Views and the Posterior)
- **One Sharpe ratio**: monthly $\widehat{SR} = 0.15$, $T = 120$, i.i.d. returns from the Normal distribution
  - annual Sharpe and its 95% interval: is it significantly positive? (Inference on One Sharpe Ratio)
- **Four tests**: $p = 0.004,\ 0.020,\ 0.030,\ 0.300$
  - Holm and Benjamini–Hochberg adjusted p-values; how many rejections at 5% under each? (Holm and Benjamini–Hochberg)
- Answers: Appendix

Check Yourself, and Next

Retrieval questions

- Why does the in-sample tangency Sharpe overstate the achievable one?
  - plug-in Sharpe; Seminar 4, A8
- By how much does it overstate it for $N = 9$, $T = 60$?
- Why does a long-only constraint act like shrinkage?
  - Jagannathan–Ma and its two-asset example
- Why can a small Sharpe difference between ERC and 1/N be nearly significant while a larger one between GMV and 1/N is not?
  - sectors; HAC test: compare the standard errors 0.023 and 0.159 in the Appendix table

Links to other chapters

- **Chapter 3**: the tangency portfolio and the CAPM
  - factor covariance models
- **Chapters 5–6**: time-varying volatility and correlation for dynamic weights
- **Chapter 7**: portfolio Value-at-Risk and Expected Shortfall
- **Chapter 13**: machine learning for returns and portfolios
  - backtest overfitting

The Frontier in Closed Form, Step by Step

- Two constraints, two multipliers: $\mathcal{L} = \frac{1}{2}\mathrm{w}^\top \boldsymbol{\Sigma}\mathrm{w} - \lambda_1(\mathrm{w}^\top \boldsymbol{\mu} - m) - \lambda_2(\mathrm{w}^\top 1 - 1)$
- First-order condition: $\boldsymbol{\Sigma}\mathrm{w} = \lambda_1\boldsymbol{\mu} + \lambda_2 1$, so $\mathrm{w} = \lambda_1\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \lambda_2\boldsymbol{\Sigma}^{-1}1$
- The constraints, with $A = 1^\top\boldsymbol{\Sigma}^{-1}1$, $B = 1^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, $C = \boldsymbol{\mu}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, $D = AC - B^2$

$$\boldsymbol{\mu}^\top \mathrm{w} = \lambda_1 C + \lambda_2 B = m, \qquad 1^\top \mathrm{w} = \lambda_1 B + \lambda_2 A = 1$$

- Solving the 2×2 system: $\lambda_1 = (Am - B)/D$, $\lambda_2 = (C - Bm)/D$
- Variance: $\mathrm{w}^\top\boldsymbol{\Sigma}\mathrm{w} = \mathrm{w}^\top(\lambda_1\boldsymbol{\mu} + \lambda_2 1) = \lambda_1 m + \lambda_2 = \dfrac{Am^2 - 2Bm + C}{D}$
- Check: at $m = B/A$ the variance is $(C - B^2/A)/D = 1/A$, the GMV
- Shapes: σ^2 is quadratic in m (a parabola in the (σ^2, m) plane)
  - $\sigma^2 - A(m - B/A)^2/D = 1/A$ is a hyperbola in the (σ, m) plane

Back

Wishart Moments and the Distribution of Weights

- i.i.d. $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ returns, maximum-likelihood estimators
  - $\hat{\boldsymbol{\mu}} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}/T)$, $T\hat{\boldsymbol{\Sigma}} \sim W_N(T-1, \boldsymbol{\Sigma})$, independent
  - Wishart: the matrix analogue of χ^2, $T-1$ degrees of freedom
- Inverse Wishart moment, finite for $T > N+2$: $\mathbb{E}[\hat{\boldsymbol{\Sigma}}^{-1}] = \frac{T}{T-N-2}\boldsymbol{\Sigma}^{-1}$
  - an **upward** bias that grows with N/T
- Conditioning on $\hat{\boldsymbol{\mu}}$ and using $\mathbb{E}[\mathrm{x}^\top \mathrm{K} \mathrm{x}] = \mathrm{tr}(\mathrm{K}\,\mathrm{Var}(\mathrm{x})) + \mathbb{E}[\mathrm{x}]^\top \mathrm{K}\,\mathbb{E}[\mathrm{x}]$
  - any random vector x, fixed matrix K, tr: trace
  - gives $\mathbb{E}[\hat{\theta}^2]$ of the lecture (Seminar 4, A8)
- GMV weights with Wishart $\hat{\boldsymbol{\Sigma}}$ and T large enough relative to N
  - $\hat{\mathrm{w}}_{\mathrm{GMV}}$ is unbiased and its exact distribution and covariance are known (Okhrin & Schmid (2006))
- Tangency weights divide by $1^\top \hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\mu}}$, which can be near zero: very heavy tails
- Estimation-risk-optimal mixtures of tangency, GMV and 1/N: Kan & Zhou (2007), Kan, Wang & Zhou (2022)

Back

Tangency Weights by Regression, Britten-Jones (1999) (1/2)

- Britten-Jones (1999): regress the constant 1 on the excess returns, **without** a constant term

$$1 = \mathrm{b}^\top \mathrm{r}_t + u_t, \qquad \hat{\mathrm{b}} = (\mathrm{X}^\top \mathrm{X})^{-1} \mathrm{X}^\top 1_T \propto \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}$$

- r_t: vector of excess returns in month t
  - X: $T \times N$ matrix with rows $\mathrm{r}_t^\top$
  - 1_T: vector of T ones; u_t: regression error
- Sherman–Morrison on $\mathrm{X}^\top \mathrm{X}/T = \hat{\boldsymbol{\Sigma}} + \hat{\boldsymbol{\mu}} \hat{\boldsymbol{\mu}}^\top$ gives the proportionality (Seminar 4, A9)
  - $\hat{\mathrm{b}}/1^\top \hat{\mathrm{b}}$ = sample tangency weights
- $1^\top \hat{\mathrm{b}}$ has the sign of $1^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}$
  - the portfolio is on the efficient branch only if it is positive

Back

Tangency Weights by Regression, Britten-Jones (1999) (2/2)

- OLS t-test of $b_i = 0$: asset i has zero weight in the tangency portfolio
  - F-test of $\mathrm{Rb} = 0$ with $\mathrm{R} = [\mathrm{I}_{N-1}, -1_{N-1}]$: "the tangency portfolio is 1/N"
- Exact finite-sample t and F distributions under i.i.d. Normal returns
  - HAC standard errors otherwise
- Nine sectors, January 1999 – July 2026 ($T = 331$): normalised $\hat{\mathrm{b}}$ reproduces the tangency weights
  - largest $|t| = 1.12$ (financials), HAC 1.19
- H_0: tangency = 1/N, eight restrictions: $F(8, 322) = 0.48$, $p = 0.87$
  - HAC Wald $\chi^2_8 = 4.50$, $p = 0.81$
  - not rejecting is not proof of equal weights

Back

The Optimal Shrinkage Intensity

- Ledoit–Wolf choose the target weight δ in $\delta \mathsf{F} + (1-\delta)\mathsf{S}$
  - to minimise the expected squared (Frobenius) distance to the true $\boldsymbol{\Sigma}$

$$\delta^* = \max\left\{0, \min\left\{1, \frac{\hat{\kappa}}{T}\right\}\right\}, \qquad \hat{\kappa} = \frac{\hat{\pi} - \hat{\rho}}{\hat{\gamma}}$$

- $\hat{\pi}$: sum of the asymptotic variances of the entries of S (noise)
- $\hat{\rho}$: their asymptotic covariances with the target entries
- $\hat{\gamma} = \|\mathsf{F} - \mathsf{S}\|^2$: misspecification of the target (not the risk aversion γ)
- Complete formulas for the constant-correlation target: Ledoit & Wolf (2004b), Appendix
  - code: the shrinkage Quantlet

Back

Shrinking the Means and Other Covariance Estimators

- Bayes–Stein (Jorion (1986)): $\hat{\boldsymbol{\mu}}_{\text{BS}} = (1 - \hat{\phi})\,\hat{\boldsymbol{\mu}} + \hat{\phi}\,\hat{\mu}_{\text{GMV}}\mathbf{1}$
  - Empirical-Bayes shrinkage of every mean towards the mean of the GMV portfolio; $\hat{\phi}$ from a Stein argument, larger when N/T is large
- Tu & Zhou (2011): the convex combination of a sophisticated rule and 1/N
  - it minimises the expected out-of-sample loss
- DeMiguel, Garlappi, Nogales & Uppal (2009): norm constraints $\|\mathrm{w}\|_1 \leq \delta$ or $\|\mathrm{w}\|_2 \leq \delta$
  - they nest the long-only GMV, shrinkage GMV and 1/N
  - the bridge to the lasso portfolios of Chapter 13
- Factor structure plus a sparse residual covariance: POET
  - principal orthogonal complement thresholding, Fan, Liao & Mincheva (2013)
  - the link to Chapter 3
- High-dimensional GMV with shrinkage towards a target portfolio: Bodnar, Parolya & Schmid (2018)

Back

Black–Litterman as Mixed (GLS) Estimation

- Theil & Goldberger (1961): stack the prior and the views as one regression on $\boldsymbol{\mu}$

$$\begin{pmatrix} \boldsymbol{\pi} \\ \mathrm{q} \end{pmatrix} = \begin{pmatrix} \mathrm{I} \\ \mathrm{P} \end{pmatrix} \boldsymbol{\mu} + \begin{pmatrix} \mathrm{e}_1 \\ \mathrm{e}_2 \end{pmatrix}, \qquad \operatorname{Var} \begin{pmatrix} \mathrm{e}_1 \\ \mathrm{e}_2 \end{pmatrix} = \begin{pmatrix} \tau\boldsymbol{\Sigma} & 0 \\ 0 & \boldsymbol{\Omega} \end{pmatrix}$$

- GLS gives exactly $\boldsymbol{\mu}_{\mathrm{BL}}$
  - with covariance $\mathrm{M} = \left[(\tau\boldsymbol{\Sigma})^{-1} + \mathrm{P}^{\top}\boldsymbol{\Omega}^{-1}\mathrm{P}\right]^{-1}$
- Equivalent form: $\boldsymbol{\mu}_{\mathrm{BL}} = \boldsymbol{\pi} + \tau\boldsymbol{\Sigma}\mathrm{P}^{\top}(\mathrm{P}\tau\boldsymbol{\Sigma}\mathrm{P}^{\top} + \boldsymbol{\Omega})^{-1}(\mathrm{q} - \mathrm{P}\boldsymbol{\pi})$
- $\boldsymbol{\Omega} \to 0$: restricted least squares, $\mathrm{P}\boldsymbol{\mu}_{\mathrm{BL}} = \mathrm{q}$
- Predictive covariance $\boldsymbol{\Sigma} + \mathrm{M}$ in the weights would change all weights
  - without views it scales them by $1/(1+\tau)$

Back

Other Risk-Based Rules

- **Inverse volatility**: $w_i \propto 1/\sigma_i$
  - equals ERC when all correlations are equal
- **Maximum diversification** (Choueifaty & Coignard (2008)): maximise $\mathrm{w}^\top \boldsymbol{\sigma} / \sqrt{\mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}}$
  - the ratio of the average asset volatility to the portfolio volatility
  - $\boldsymbol{\sigma}$: vector of asset volatilities
- **Minimum variance**: GMV, the extreme risk-based rule
- All ignore expected returns: robust to estimation error
  - but optimal only under special assumptions

Back

Results, Sector Universe

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown	Sharpe, 50 bp
1/N	9.4%	14.3%	0.66	0.02	−49%	0.65
MV	17.8%	69.0%	0.26	3.86	−100%	−0.08
MV-LO	10.7%	14.7%	0.72	0.23	−42%	0.63
GMV	8.5%	11.9%	0.71	0.23	−33%	0.59
GMV-LO	8.1%	11.5%	0.70	0.09	−34%	0.65
GMV-LW	7.6%	11.8%	0.65	0.14	−31%	0.58
ERC	9.3%	13.2%	0.70	0.03	−47%	0.69
HRP	8.9%	12.8%	0.70	0.08	−45%	0.66

- January 2004 – July 2026
  - turnover: monthly mean
  - drawdown of total-return wealth
- Last column: net of 50 basis points per unit of turnover

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Results, Multi-Asset Universe (8 ETFs)

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown	Sharpe, 50 bp
1/N	4.3%	8.4%	0.52	0.02	−21%	0.50
MV	−113.0%	517.5%	−0.22	44.46	−100%	−0.54
MV-LO	7.1%	8.9%	0.80	0.12	−24%	0.72
GMV	1.4%	3.8%	0.37	0.14	−11%	0.16
GMV-LO	1.7%	5.4%	0.32	0.05	−17%	0.27
GMV-LW	1.1%	4.2%	0.27	0.09	−13%	0.15
ERC	3.6%	7.3%	0.49	0.02	−21%	0.47
HRP	2.4%	6.6%	0.36	0.04	−20%	0.32

- May 2012 – July 2026 (171 months)
  - MV's mean and volatility are arithmetic averages of monthly returns that include a loss above 100%, so only its drawdown has an economic meaning

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Results, Combined Universe (16 ETFs)

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown	Sharpe, 50 bp
1/N	7.9%	10.3%	0.76	0.02	−17%	0.75
MV	28.7%	261.8%	0.11	22.10	−100%	−0.51
MV-LO	8.0%	9.8%	0.81	0.20	−26%	0.69
GMV	0.8%	4.1%	0.20	0.24	−12%	−0.15
GMV-LO	2.0%	5.4%	0.36	0.06	−17%	0.30
GMV-LW	0.9%	4.4%	0.20	0.13	−14%	0.03
ERC	5.5%	7.8%	0.71	0.03	−18%	0.68
HRP	4.0%	7.0%	0.56	0.11	−18%	0.47

- May 2012 – July 2026

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All 21 Tests against 1/N (1/3), Sectors

Rule	$\hat{\Delta}$	HAC SE	p HAC	Holm	p one-sided	p bootstrap	Block
MV	−0.40	0.331	0.22	1.00	0.889	0.30	6
MV-LO	+0.06	0.120	0.60	1.00	0.299	0.63	1
GMV	+0.05	0.159	0.75	1.00	0.377	0.76	1
GMV-LO	+0.04	0.112	0.73	1.00	0.363	0.73	1
GMV-LW	−0.02	0.166	0.93	1.00	0.537	0.94	1
ERC	+0.04	0.023	0.09	1.00	0.046	0.13	1
HRP	+0.04	0.040	0.32	1.00	0.162	0.36	1

- January 2004 – July 2026, $T = 271$
  - $\hat{\Delta}$: annualised Sharpe of the rule minus that of 1/N, before costs
- HAC: Bartlett, 4 lags, two-sided
  - Holm over all 21 two-sided HAC tests
  - one-sided: $1 - \Phi(\hat{\Delta}/\mathrm{SE})$
  - bootstrap: studentized circular block bootstrap, two-sided, block from Algorithm 3.1

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All 21 Tests against 1/N (2/3), Multi-Asset

Rule	$\hat{\Delta}$	HAC SE	p HAC	Holm	p one-sided	p bootstrap	Block
MV	−0.74	0.300	0.01	0.29	0.993	0.06	10
MV-LO	+0.29	0.184	0.12	1.00	0.059	0.16	10
GMV	−0.15	0.226	0.51	1.00	0.745	0.54	1
GMV-LO	−0.20	0.137	0.14	1.00	0.930	0.15	8
GMV-LW	−0.25	0.191	0.19	1.00	0.903	0.20	4
ERC	−0.03	0.060	0.61	1.00	0.694	0.58	10
HRP	−0.16	0.089	0.07	1.00	0.964	0.05	6

- May 2012 – July 2026, $T = 171$
  - same columns as for the sectors

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All 21 Tests against 1/N (3/3), Combined

Rule	$\hat{\Delta}$	HAC SE	p HAC	Holm	p one-sided	p bootstrap	Block
MV	−0.66	0.355	0.06	1.00	0.968	0.22	10
MV-LO	+0.05	0.210	0.82	1.00	0.409	0.82	10
GMV	−0.56	0.261	0.03	0.58	0.985	0.05	1
GMV-LO	−0.40	0.203	0.05	0.86	0.976	0.08	6
GMV-LW	−0.56	0.249	0.02	0.48	0.988	0.03	2
ERC	−0.06	0.114	0.61	1.00	0.695	0.62	10
HRP	−0.20	0.152	0.18	1.00	0.908	0.20	8

- May 2012 – July 2026, $T = 171$
- One-sided family: Holm $21 \times 0.046 = 0.96$
  - Benjamini–Hochberg $\min\{21 \times 0.046,\ \frac{21}{2} \times 0.059\} = 0.62$ for the two smallest

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Bootstrap Details and the Block Length

- Studentization: $s(\hat{\Delta}^*)$ from the block sums of each resample
  - $s(\hat{\Delta})$ on the data from a prewhitened quadratic-spectral kernel (Andrews & Monahan (1992)) with automatic bandwidth (Andrews (1991))
- Ledoit & Wolf (2008), Algorithm 3.1 (grid and K of Sections 4–5, applied to each of the 21 pairs)
  - choose the block b whose bootstrap interval has actual coverage closest to 95%
  - Fit a first-order vector autoregression to the pair of returns; resample its residuals by a stationary bootstrap (Politis & Romano (1994)), mean block 5
  - Generate $K = 5000$ pseudo-samples of length T; in each, build the 95% interval for $b \in \{1, 2, 4, 6, 8, 10\}$
  - $\hat{g}(b)$: share of intervals that contain the pseudo-true $\hat{\Delta}$; pick $\tilde{b} = \arg\min_b |\hat{g}(b) - 0.95|$
- A non-studentized bootstrap with a fixed 6-month block is not this test
  - its block is not calibrated for coverage

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The Deflated Sharpe Ratio

- Bailey & López de Prado (2014): the best of N Sharpe ratios is biased upwards even if every true Sharpe ratio is zero
- Expected maximum under H_0: all true Sharpe ratios zero, N independent trials
  - monthly units as T
  - on this slide N counts trials (configurations), not assets
  - $\gamma_E = 0.5772$: Euler's constant; $\hat{V}$: the variance of the N estimates
  - Φ: standard Normal distribution function; e: Euler's number

$$SR_0 = \sqrt{\hat{V}}\left[(1-\gamma_E)\,\Phi^{-1}\left(1-\tfrac{1}{N}\right) + \gamma_E\,\Phi^{-1}\left(1-\tfrac{1}{Ne}\right)\right]$$

- Deflated Sharpe ratio: the probability that the true Sharpe of the winner exceeds SR_0
  - with skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$ of the winner

$$\text{DSR} = \Phi\left(\frac{(\widehat{SR} - SR_0)\sqrt{T-1}}{\sqrt{1-\hat{\gamma}_3\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{SR}^2}}\right)$$

- This chapter: 24 configurations (8 rules $\times$ 3 universes); winner MV-LO on 16 ETFs, 0.81 a year
  - As 24 independent trials: $SR_0 = 0.52$ a year, DSR $= 0.84 < 0.95$
  - The configurations are strongly correlated (up to 0.99), so fewer are independent: $N = 8$ gives DSR 0.93, $N = 3$ gives 0.98
- Holm adjusts the whole family of pairwise tests against 1/N
  - the DSR asks whether the single best configuration is better than luck

Back

Beyond Equities, Digital Assets

- Petukhina, Trimborn, Härdle & Elendner (2021): cryptocurrencies in portfolio allocation rules
  - gains depend on the rule and on liquidity
- Pele et al. (2023): are cryptos becoming alternative assets?
- Bitcoin: annualised volatility 172%
  - monthly, 2010–2026, one surviving coin chosen ex post
  - for comparison, IEF 6.6%
- Risk contributions grow with volatility
  - risk-based rules give Bitcoin a small weight automatically (Seminar 4, B8)
- Robust mean–variance for crypto-assets: Pele (2023)

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Case Study, Method in Detail (1/2)

- **Cost** (eqs. 4, 7)
  - π_t: risky-asset weights; $1 - 1^\top \pi_t$: the weight of the risk-free asset
  - $g_t = \mathrm{diag}\big((1 + r_t^f + r_t)/(1 + g_t^w)\big)$ (eq. 6); r_t^f: risk-free rate, r_t: asset returns, g_t^w: growth of wealth
  - $\mathbf{\Lambda} = \mathrm{diag}(0.2/V_i)$; V_i: average daily dollar volume over 6 months; trading 1% of V_i moves the price by 0.1% (eq. 35, Frazzini, Israel & Moskowitz (2018))
- **Optimal rule** (Proposition 3): $\pi_t = m\, g_t \pi_{t-\mathbf{1}} + (\mathbf{I} - m) A_t$, generalising Gârleanu & Pedersen (2013)
  - A_t: a weighted sum of today's and expected future Markowitz portfolios; m solves eq. (14)
- **Portfolio-ML** (Proposition 4, eq. 40): $A_t = \mathrm{diag}(1/\sigma_{i,t})\, f(s_t)\beta$
  - $\sigma_{i,t}$: volatility of asset i; f: p random Fourier features of the signals s_t
  - $\hat{\beta} = (E_T[\tilde{\mathbf{\Sigma}}_t] + \lambda \mathbf{I})^{-\mathbf{1}} E_T[\tilde{r}_{t+\mathbf{1}}]$: a ridge regression ($\lambda$: penalty) that maximises utility net of costs in closed form (eqs. 24–26)
  - it learns the weights directly, as Brandt, Santa-Clara & Valkanov (2009) do without costs

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Case Study, Method in Detail (2/2)

- **Data** (Section 5.1): US common stocks with market capitalisation above the NYSE median (184–805 stocks)
  - 115 characteristics of Jensen, Kelly & Pedersen (2023), ranked into $[0, 1]$
  - December 1952 – December 2020, out of sample 1981–2020, monthly
- **Covariance $\boldsymbol{\Sigma}$**: 13 characteristic themes and 12 industries as factors
  - exponential weights with half-lives of 378 days (correlations) and 126 days (variances)
  - investor $\gamma = 10$, wealth \$10 billion at the end of 2020, growing with the market (Section 5.1.3)
- **Tuning** (Table 1), chosen each year by the utility of validation backtests since 1971
  - $\lambda \in \{0, e^{-\mathbf{10}}, e^{-\mathbf{9.8}}, \ldots, e^{\mathbf{10}}\}$, $p \in \{2^{\mathbf{6}}, \ldots, 2^{\mathbf{9}}\}$, $\eta \in \{e^{-\mathbf{3}}, e^{-\mathbf{2}}\}$; η: standard deviation of the random weights
- **Rivals** ("*" marks an extra tuning layer, Section 5.3)
  - Markowitz-ML (eq. 32), Static-ML (one-period cost, eq. 34), Multiperiod-ML (forecasts for 12 horizons, eq. 22)
  - factor sorts, 1/N, minimum variance, market

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Case Study, Complete Results

- **Table 2**: Portfolio-ML turnover 0.25 a month
- Portfolio-ML needs no second tuning layer, unlike Static-ML* and Multiperiod-ML*
- **Table 3**: posterior probability that Portfolio-ML has the higher utility
  - 99.9% against Static-ML* and Multiperiod-ML*, 100% against Markowitz-ML
- **Figure 1(B)** (Section 6.2), $\gamma = 10$: size is a cost
  - with \$10 billion, net excess return 14.6% at 11% volatility; with \$1 billion, 16.3%
- **Figure 5**: net of costs, quality is the most important theme
  - short-term reversal is almost irrelevant
  - monthly autocorrelation -0.04 for 1-month reversal, 0.94 for book-to-market (Section 6.3)
  - persistent signals are cheap to trade
- **Limits stated by the authors**
  - fees and other frictions; computing power and ML methods not available in real time early in the sample (Section 6.1)
  - costs and risk treated as constant over the horizon of each optimisation (Section 5.1.3)
  - among the smallest 20% of stocks, Portfolio-ML ranks behind Multiperiod-ML* and Static-ML* (Figure 8)

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Case Study, the Course Replication (1/2)

- **Course data**: 14 US stocks and 9 sector ETFs
  - five signals: 1-month log return, log momentum from $t-6$ and from $t-12$ to $t-1$, 1-year volatility, 6-month dollar volume (formulas in Seminar 4, C3)
  - cost rule, γ, wealth and grid as in the paper; $\boldsymbol{\Sigma}$ with the same exponential weights; validation from 2011
- **Portfolio-ML**: cost 0.6% a year, leverage 0.32
- Net utility: Static-ML -0.13, minimum variance -0.72
  - Markowitz-ML extremely negative (leverage 18)
- **Posterior probability** of a higher utility for Portfolio-ML
  - 99.5% against Static-ML, 99.9% against minimum variance, 93.3% against 1/N

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Case Study, the Course Replication (2/2)

- Same ranking of cost-aware against cost-agnostic rules as Table 2
  - Portfolio-ML wins with small, slowly traded positions, not with a higher gross Sharpe ratio (0.68 against 0.87 for 1/N)
- **Limits**
  - 23 assets instead of 184–805; five signals instead of 115
  - no factor model for $\mathbf{\Sigma}$, whose window reaches 10 years only in August 2014 (common prices from August 2004)
  - return forecasts with $p \geq 2^6$ and three validation years instead of ten; no starred rivals
- **Open extension** (Seminar 4, C3)
  - implementable frontiers for wealth of \$1, 10 and 100 billion and $\gamma \in \{1, 5, 10, 20, 100\}$
  - 1/N's trading costs computed at each wealth level

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The Discovery Loop, Reusable Prompts (1/2)

- Tools are interchangeable
  - an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot)
  - literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature**
  - *"List peer-reviewed studies since 2015 comparing shrinkage or risk-based rules with 1/N out of sample; for each give DOI, N, T, cost assumption and test."*
  - Resolve every DOI on Crossref and open the table cited
- **Hypotheses**
  - *"Give three competing explanations for a Sharpe gain over 1/N (better Σ, a low-volatility premium, luck) and the testable implication of each for turnover, subperiods and factor loadings."*
- **Data and code**
  - *"Write Python that reads the course's daily price files, aligns prices on common dates and runs a 60-month rolling GMV-LW and 1/N; weights at t may use data up to $t-1$ only."*
  - Replicate first: 1/N 0.76, GMV-LW 0.20 on the combined universe

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The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**
  - *"Draft a pre-registration: the family of tests (rules × universes × windows × costs), the error rate (FWER or FDR), the subperiods and a placebo, fixed before any result is seen."*
- **Critique**
  - *"Act as a referee: give the three strongest reasons my Sharpe gain is spurious and the test that would expose each."*
- **Report**: every specification tried, adjusted p-values, the power of the design, code and seeds
  - any specification chosen after seeing results is exploratory

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Answers to the Quick Checks

- **Three rules**: $s_{11} = 0.09$, $s_{22} = 0.0225$, $s_{12} = 0.2 \times 0.30 \times 0.15 = 0.009$
  - GMV: $w_1 = (0.0225 - 0.009)/(0.09 + 0.0225 - 0.018) = 1/7$, volatility 14.3%; ERC (inverse volatility): $w_1 = 1/3$, volatility 15.5%; 1/N: 18.1%
  - 1/N: $w_1(\Sigma w)_1 = 0.5 \times 0.0495 = 0.02475$ of $\sigma_p^2 = 0.0326$: asset 1 carries 76% of the risk with 50% of the weight
- **Plug-in Sharpe**: monthly $\theta^2 = 0.25/12 = 0.0208$
  - $(120 \times 0.0208 + 4)/(120 - 6) = 0.0570$; $\sqrt{12 \times 0.0570} = 0.83$ a year, against a true 0.50
- **Noise band**: $c = 10/40 = 0.25$, band $[(1 - 0.5)^2, (1 + 0.5)^2] = [0.25, 2.25]$
  - 2.0 lies inside, so this comparison cannot separate it from noise
- **One view**: equal precisions give view weight $0.0009/(0.0009 + 0.0009) = 0.5$, posterior 7%
  - with $\Omega = 0.0003$, weight $0.0009/0.0012 = 0.75$, posterior $4\% + 0.75 \times 6\% = 8.5\%$
- **One Sharpe ratio**: SE $= \sqrt{(1 + 0.15^2/2)/120} = 0.092$
  - annual $0.52 \pm 1.96 \times 0.32$, interval $[-0.10, 1.14]$: not significant
- **Four tests**
  - Holm 0.016, 0.060, 0.060, 0.300: one rejection
  - Benjamini–Hochberg 0.016, 0.040, 0.040, 0.300: three rejections

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Courses, Code and Data

Advanced courses on quantinar.com

- Hierarchical Clustering
- Minimum Spanning Tree
- Multivariate Statistical Analysis

Code and data

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_04
- Risk-free rate: Kenneth French Data Library

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Acronyms used in this material (1/3)

- **AI**: Artificial Intelligence
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAPM**: Capital Asset Pricing Model
- **DSR**: Deflated Sharpe Ratio
- **ERC**: Equal Risk Contribution portfolio
- **ETF**: Exchange-Traded Fund
- **FDR**: False Discovery Rate
- **FWER**: Family-Wise Error Rate
- **GLS**: Generalised Least Squares
- **GMV**: Global Minimum-Variance portfolio
- **GMV-LO**: Global Minimum-Variance portfolio, Long-Only
- **GMV-LW**: Global Minimum-Variance portfolio with Ledoit–Wolf covariance
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HRP**: Hierarchical Risk Parity
- **HYG**: iShares iBoxx High Yield Corporate Bond ETF
- **IEF**: iShares 7–10 Year Treasury Bond ETF
- **LO**: Long-Only, no short sales
- **LQD**: iShares iBoxx Investment Grade Corporate Bond ETF
- **ML**: Machine Learning
- **MV**: Mean–Variance tangency portfolio estimated from sample moments
- **MV-LO**: Mean–Variance tangency portfolio, Long-Only
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **OMV**: Österreichische Mineralölverwaltung — *Austrian Mineral Oil Administration (OMV group)*
- **OOS**: Out-Of-Sample — *evaluation on data after the estimation period*
- **POET**: Principal Orthogonal complEment Thresholding
- **US**: United States

Acronyms used in this material (2/3)

- **BH**: Benjamini–Hochberg (procedure controlling the FDR)
- **DOI**: Digital Object Identifier
- **i.i.d.**: independent and identically distributed
- **LW**: Ledoit–Wolf (shrinkage covariance estimator)
- **QS**: Quadratic Spectral (kernel)
- **RON**: Romanian leu
- **SE**: Standard Error
- **SR**: Sharpe Ratio
- **USD**: US dollar

Acronyms used in this material (3/3): asset tickers

- **SPY**: SPDR S&P 500 ETF Trust
- **EFA**: iShares MSCI EAFE ETF (developed markets outside the US and Canada)
- **EEM**: iShares MSCI Emerging Markets ETF
- **TLT**: iShares 20+ Year Treasury Bond ETF
- **GLD**: SPDR Gold Shares
- **XLB**, **XLE**, **XLF**, **XLI**, **XLK**: Select Sector SPDR (materials, energy, financials, industrials, technology)
- **XLP**, **XLU**, **XLV**, **XLY**: Select Sector SPDR (consumer staples, utilities, health care, consumer discretionary)
- **TLV**: Banca Transilvania
- **SNP**: OMV Petrom
- **BRD**: BRD–Groupe Société Générale
- **TGN**: Transgaz
- **SNG**: Romgaz
- **SNN**: Nuclearelectrica
- **EL**: Electrica
- **TEL**: Transelectrica